

WinRiver II

Section-By-Section

Quick Start Guide

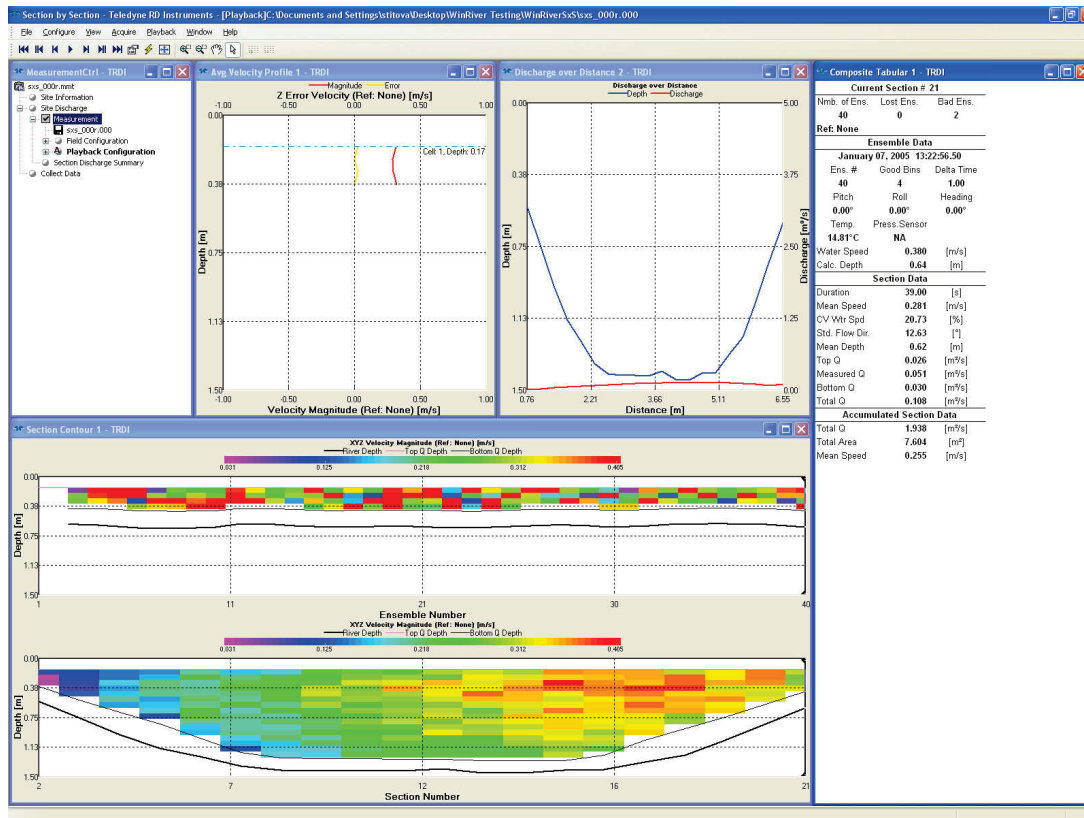


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WinRiver II Section-by-Section Quick Start Guide

Introduction

Thank you for purchasing the Teledyne RD Instruments (TRDI) *WinRiver II Section-By-Section (SxS)* software. This Quick Start Guide will lead you through the steps required for a successful river discharge measurement. Please read the entire guide, and then follow the instructions in the order they are presented. Additional information can be found in the *WinRiver II Section-By-Section* User's Guide that is supplied on CD-ROM.



NOTE. To purchase a printed copy of the *WinRiver II* Section-By-Section User's Guide, contact our Customer Service department at rdifs@teledyne.com or call +1 (858) 842-2600 and order the *WinRiver II* Section-By-Section User's Guide P/N 957-6237-00.



NOTE. This guide covers *WinRiver II* Section-By-Section version 2.07.

How to Contact Teledyne RD Instruments

If you have technical issues or questions involving a specific application or deployment with your instrument, contact our Field Service group:

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Overview

The first step is to become familiar with the hardware and software that comes with the ADCP.

This Section Covers:

- Comparison of WinRiver II and Section-By-Section
- Hardware Overview
- Software Overview
- Installing the Software

Comparison of WinRiver II and Section-By-Section

You will notice several similarities between *WinRiver II* and *Section-By-Section* and also several differences.

- Many of the displays are similar but some are unique to *Section-By-Section*. Note that when you are in *Section-By-Section*, the title bar at the top of the display is **Section-By-Section**. This makes it easy to see which program is running.
- In *WinRiver II*, the transect is done by crossing from bank to bank of the river. In *Section-By-Section*, transects are done by measuring at different user selected points on the river. This makes it possible to measure streams that are frozen over. This makes the **Discharge Summary** display different. *WinRiver II* shows the history of the entire transect; *Section-By-Section* shows the history of each section measured.
- Measurement files are different between *WinRiver II* and *Section-By-Section*. Use the proper program to play back data. A measurement data file created with *Section-By-Section* can not be used to play back data using *WinRiver II*.
- *WinRiver II* uses Bottom Tracking or GPS for velocity reference and *Section-By-Section* does not. This makes it possible to use *Section-By-Section* software to measure discharge in streams with moving beds.

Hardware Overview

Take a moment to familiarize yourself with the requirements for a River Discharge measurement.

ADCP Requirements

In order to use *WinRiver II Section-By-Section*, your Workhorse ADCP must meet the following criteria.

- WorkHorse Rio Grande ADCP
- or
- The WorkHorse ADCP **must** have the Bottom Track upgrade installed



NOTE. The Bottom Track and High Resolution Water Modes upgrades are available for WorkHorse Monitor and Sentinel ADCPs (included with the Rio Grande ADCP). *This guide assumes you will be using a WorkHorse Rio Grande ADCP.*

The Rio Grande, Monitor / Sentinel, and Mariner ADCPs can benefit from the upgrade to the High Rate Pinging (Water Mode 12) and Shallow Bottom Mode (Bottom Mode 7). Contact your local sales representative if you are interested in upgrading your system.

- *WinRiver II Section-By-Section* can also be used with a RiverRay and StreamPro ADCP if your computer has a Bluetooth connection.



NOTE. In StreamPro Firmware version 31.07 the Long Range mode has been extended to 6 meters. For compatibility with WinRiver II v2.06 and above, existing Long Range StreamPro users must have their firmware updated to version 31.07 (or higher).

ADCP Mounting Requirements

The ADCP must be mounted as follows.

- Over the side or through the hull mounted
- Downward facing
- Within 5 degrees of vertical

TRDI recommends using a fixed mount to achieve the best performance (see the ADCP Technical Manual for recommended sources of mounts).

Installing the Software

You will be installing two software packages: *RDI Tools* (contains *BBTalk*) and *WinRiver II*. Installing *WinRiver II* will also install *WinRiver II Section-By-Section*. These programs will be required for testing and measurements.

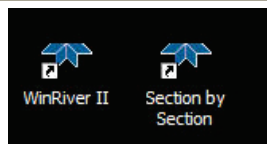
The software requires the following:

- Windows Vista®, Windows XP Pro® or Windows 2000®
- Pentium III 600 MHz class PC (higher recommended)
- 64 megabytes of RAM (128 MB RAM recommended)
- 50 MB Free Disk Space plus space for data files (A large, fast hard disk is recommended)
- One Serial Port (two or more High Speed UART Serial Ports recommended)
- Minimum display resolution of 1024 x 768
- CD-ROM Drive
- Mouse or other pointing device



Software Installation

- Insert the compact disc into your CD-ROM drive and then follow the browser instructions on your screen. If the browser does not appear, complete Steps "b" through "d."
- On the Windows task bar, click the **Start** button, and then click **Run**.
- Type <drive>:launch. For example, if your CD-ROM drive is drive D, type d:launch.
- Follow the browser instructions on your screen



Click the **Section-By-Section** icon to start *Section-By-Section*.

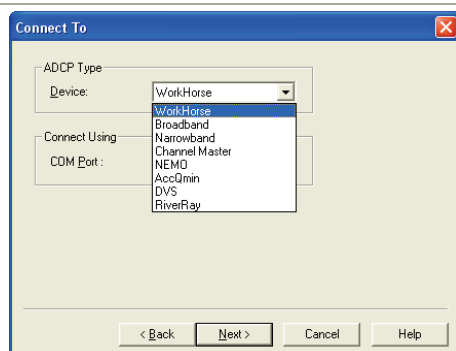
Click the **WinRiver II** icon to start *WinRiver II* (See the WinRiver II Quick Start Guide for details on how to use WinRiver II).

Setting up Communications

The software needs to have the same configuration as the ADCP / Radio Modems to enable any form of communication.

BBTalk Communications Setup

Use these next steps to confirm communications with the ADCP. See [Commonly Used BBTalk Commands](#) for commonly used commands sent via *BBTalk*. The *BBTalk* software assumes the last communication settings are correct and opens. However, the first time the software is run it is necessary to setup the communications. Therefore, on the first start up, do the following.



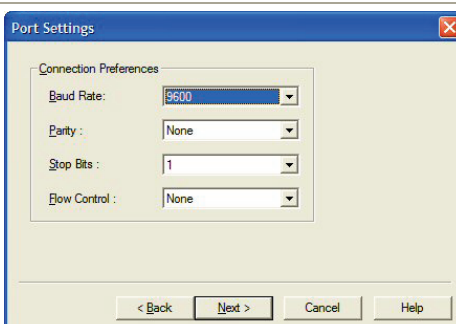
Start BBTalk

Start the *BBTalk* program.

On the **Connect To** screen, select **WorkHorse**.

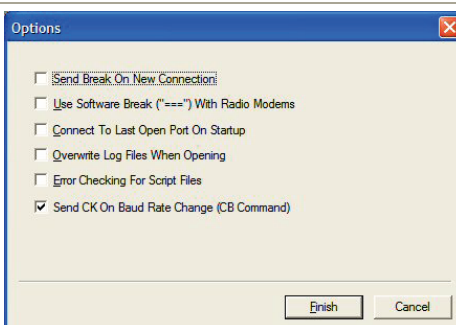
Select the COM port the ADCP cable is connected to.

Click **Next**.



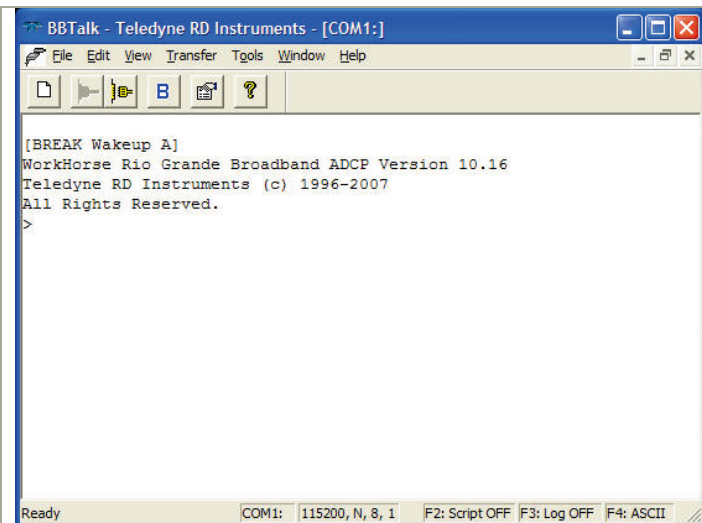
Enter the **Baud Rate**, **Parity**, **Stop Bits**, and **Flow Control**. If you are unsure of the settings, leave them at the default settings.

Click **Next**.



Un-check all boxes except for the **Send CK on Baud Rate Change (CB Command)** box.

Click **Finish**.

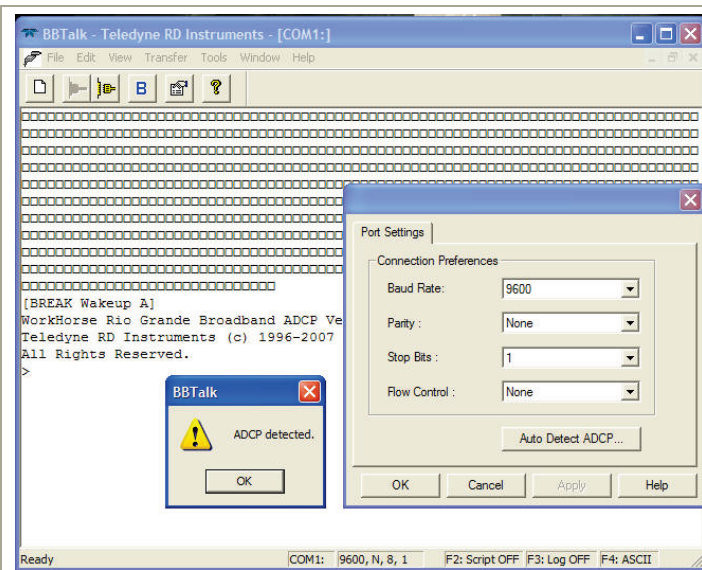


Wakeup

On the **File** menu, click **Break** (you can also press the **End** key to send a break or and press the **B** button on the Tool-bar).

To send a break when using a Radio Modem, press === (3 equals).

You should see the wakeup message appear on the log file window.



If the wakeup message is not readable, do the following.

On the **File** menu, click **Properties**.

Click the **Auto Detect ADCP** button.

Click **OK** when the ADCP is detected.

Try to wakeup the ADCP again.

What if the ADCP does not respond?

If your ADCP does not respond, check the serial port, cables, AC power, 12 VDC battery power (Rio Grande ADCPs), and battery pack (StreamPro ADCPs). If necessary, refer to the Troubleshooting section in the ADCP Technical Manual.

Changing the Baud Rate in the ADCPs

The ADCP can be set to communicate at baud rates from 300 to 115200. The factory default baud rate is always 9600 baud. The baud rate is controlled via the CB-command. The following procedure explains how to set the baud rate and save it in the ADCP. This procedure assumes that you will be using the program *BBTalk* that is supplied by Teledyne RD Instruments.



NOTE. This applies to WorkHorse ADCPs only. StreamPro and RiverRay ADCPs use a fixed baud rate (see [StreamPro and RiverRay Communication Setup](#)).

```
>
[BREAK Wakeup A]
WorkHorse Rio Grande Broadband ADCP Version 10.15
RD Instruments (c) 1996-2005
All Rights Reserved.
>
>cr1
[Parameters set to FACTORY defaults]
```

Connect the ADCP to the computer and apply power.

Start the *BBTalk* program and establish communications with the ADCP. Wakeup the ADCP by sending a break signal with the **End** key.

At the ">" prompt in the communication window, type **CR1** then press the Enter key. This will set the ADCP to the factory default settings.

BAUD RATE	CB-command
300	CB011
1200	CB111
2400	CB211
4800	CB311
9600	CB411 (Default)
19200	CB511
38400	CB611
57600	CB711
115200	CB811

Send the CB-command that selects the baud rate you wish. The table on the left shows the CB-command settings for different baud rates with no parity and 1 stop bit.

For example, to change the baud rate to 115200, at the ">" prompt in the communication window, type **cb811** then press the Enter key.



The CB? command will identify the communication setting.

```
>cb?
CB = 411 ----- Serial Port Control (Baud
[4=9600]; Par; Stop)
>cb811
>CK
[Parameters saved as USER defaults]
>cb?
CB = 811 ----- Serial Port Control (Baud
[8=115200]; Par; Stop)
>
```

BBTalk will send the command **CK** to save the new baud rate setting.

Exit *BBTalk*.

The ADCP is now set for the new baud rate. The baud rate will stay at this setting until you change it back with the CB-command.

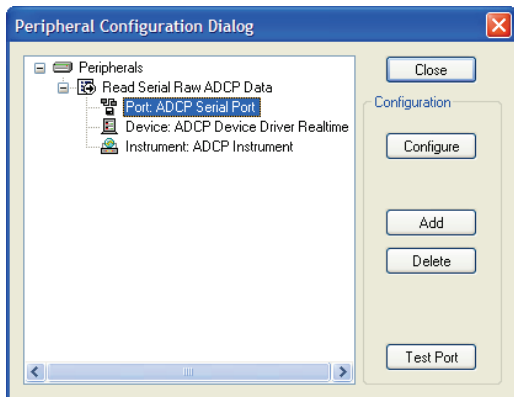


Exit *BBTalk* so the communication port is available for use with *WinRiver II* SxS.

Section-By-Section Communications Setup

When *WinRiver II SxS* is first started, you must set up the communications with the ADCP. Once setup, *WinRiver II SxS* will remember the settings and use them each time the program is started.

ADCP Communications Setup



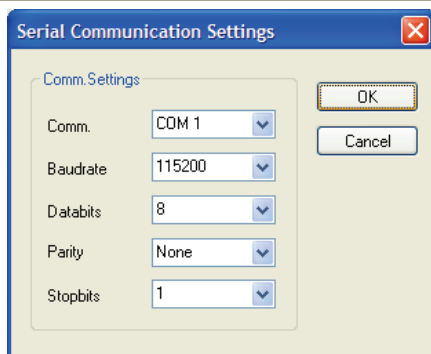
Connect and power up the ADCP as shown in the appropriate ADCP User's Guide.

Start *WinRiver II SxS*.

On the **Configure** menu, select **Peripherals**.

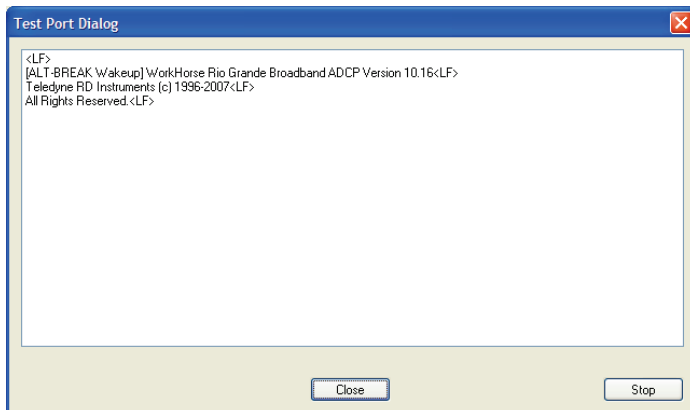
Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Port: ADCP Serial Port**.

Press the **Configure** button.



Select the communication port, Baudrate, Databits, Parity, and Stopbits you set using *BBTalk* (see [Changing the Baud Rate in the ADCPs](#)).

Click **OK** to return to the **Peripheral Configuration Dialog**.



Click the **Test Port** button to connect to the ADCP and confirm the communication setting.

You should see the wakeup message appear on the log file window.

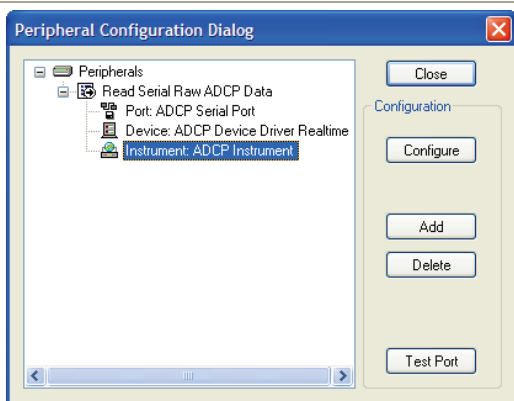
Click the **Close** button to return to the **Peripheral Configuration Dialog**.



If no wakeup message appears, see [Advanced ADCP Communications Configuration](#) and use the **Send Break** button.

Advanced ADCP Communications Configuration

Use the following steps to communicate with the ADCP or verify command settings.



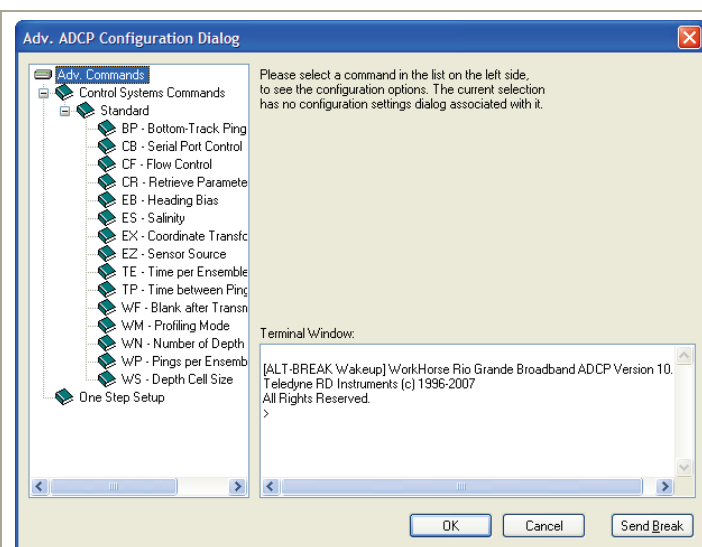
On the **Configure** menu, select **Peripherals**.

Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Instrument: ADCP Instrument**.

Press the **Configure** button.



Right-click on **Instrument: ADCP Instrument** and select **Auto Detect** to automatically detect the serial port settings.



Use the **Advanced ADCP Configuration Dialog** screen to send or verify command setting as needed.

If the command you want to send the ADCP is not on the list, type it next to the ">" prompt in the **Terminal Window** and press return.



Use the **Send Break** button to wake up the ADCP.

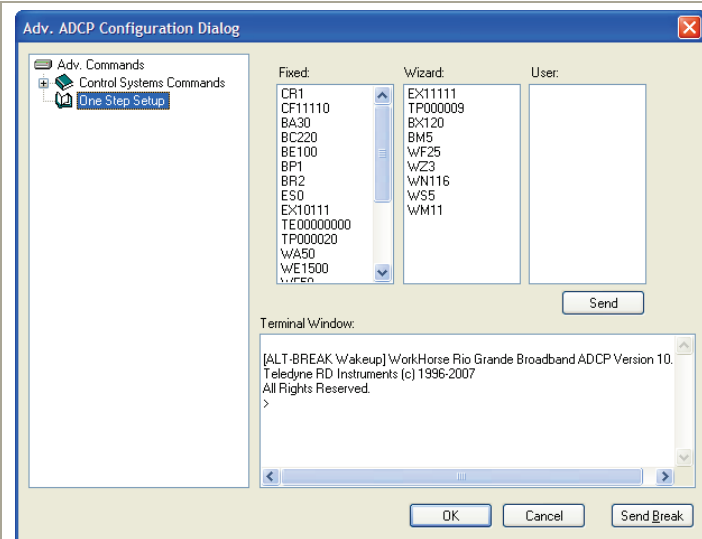


CAUTION. Do not use this screen to deploy or configure the ADCP. The commands will be overwritten when **Acquire, Start Pinging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pinging (F4)** can be used to deploy the ADCP.



NOTE. WorkHorse Monitor/Sentinel ADCPs will go to sleep if a command is not sent within five minutes; Use the **Send Break** button to wake the ADCP before you start pinging.

WorkHorse Rio Grande ADCPs do not sleep. For more information, see the CL command in the WorkHorse Commands and Output Data Format guide.



Use the **One Step Setup** to check the deployment commands sent to the ADCP after the **Measurement Wizard** has been completed (see [Run the Measurement Wizard](#)).

Run the Measurement Wizard.

Click **One Step Setup**. Click the **Send** button.

If any command generates an error message, stop and correct the problem before deploying.

 This is a good way to check if any command generates an error message.



CAUTION. Do not use this screen to deploy or configure the ADCP. The commands will be overwritten when **Acquire, Start Pinging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pinging (F4)** can be used to deploy the ADCP.



NOTE. The **One Step Setup** is a good way to check if any command generates an error message (for example, if the commands depend on the High Resolution Water Profiling upgrade but you are not sure the ADCP has this feature).

If a command generates an error, a message box will appear.

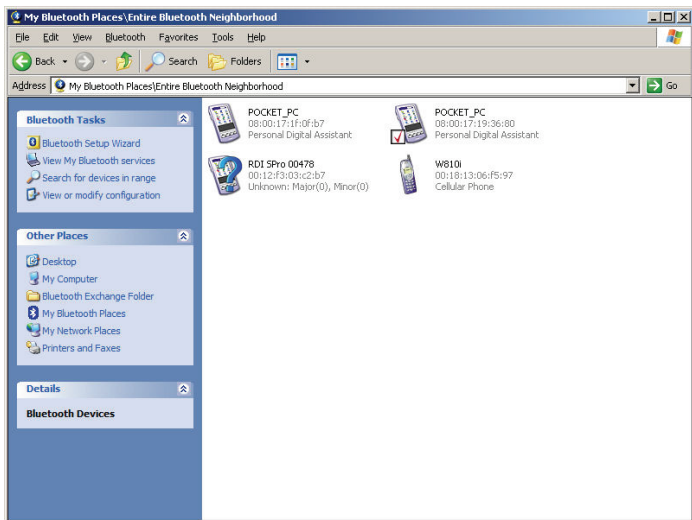
The **One Step Setup** does not check the commands for “reasonableness”.

If the wizard has not been run, the **Fixed** and **Wizard** command boxes will be blank.

Commands can be entered in the **User** box or at the “>” prompt.

StreamPro and RiverRay Communication Setup

WinRiver II SxS can also be used with a RiverRay and StreamPro ADCP if your computer has a Bluetooth connection.



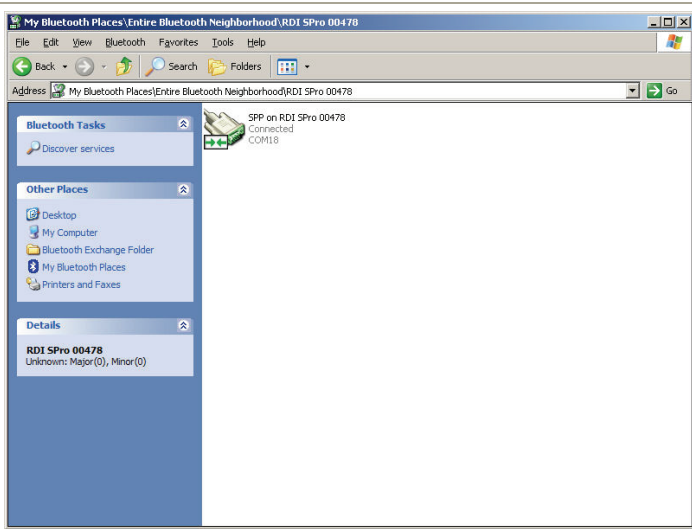
The screenshot shows the 'My Bluetooth Places\Entire Bluetooth Neighborhood' window. On the left, there are sections for 'Bluetooth Tasks' (including Bluetooth Setup Wizard, View My Bluetooth services, Search for devices in range, and View or modify configuration), 'Other Places' (Desktop, My Computer, Bluetooth Exchange Folder, My Bluetooth Places, My Network Places, Printers and Faxes), and 'Details' (Bluetooth Devices). The main area displays four discovered devices: POCKET_PC (08:00:17:1F:0F:b7, Personal Digital Assistant), POCKET_PC (08:00:17:19:36:80, Personal Digital Assistant), RDI SPro 00478 (00:12:f3:03:c2:b7, Unknown: Major(0), Minor(0)), and W810i (00:18:13:06:f5:97, Cellular Phone).

Turn on the power to the ADCP.

Configure your Bluetooth device. It should locate the RiverRay or StreamPro ADCP (for example **RDI SPro 00xxx** or **RDI RRay 00xxx** for a River-Ray system where xxx is the serial number).

 This screen may look different based on what Bluetooth device you are using.

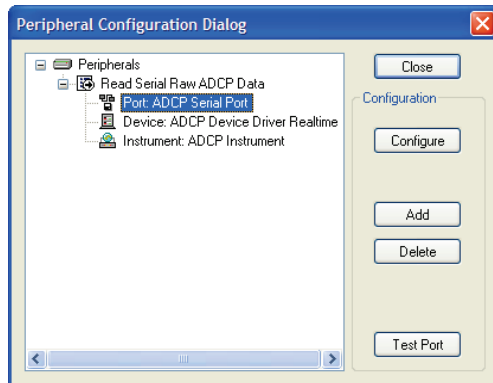
 Some Bluetooth configurations require a pin code for StreamPro and RiverRay ADCPs. The pin code is 0.



The screenshot shows the 'My Bluetooth Places\Entire Bluetooth Neighborhood\RDI SPro 00478' window. The left sidebar is the same as the previous screenshot. The main area shows a single device, 'SPP on RDI SPro 00478', which is 'Connected' and uses 'COM18'.

Double-click on the **RDI SPro** icon and make sure that that it shows Connected.

Make note of the **Com Port** number it is using. On this example, the StreamPro is connected to Com Port 18.

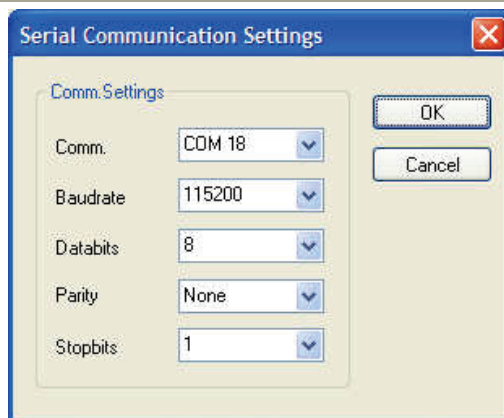


Start *WinRiver II SxS*.

On the **Configure** menu, select **Peripherals**.

Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Port: ADCP Serial Port**.

Press the **Configure** button.

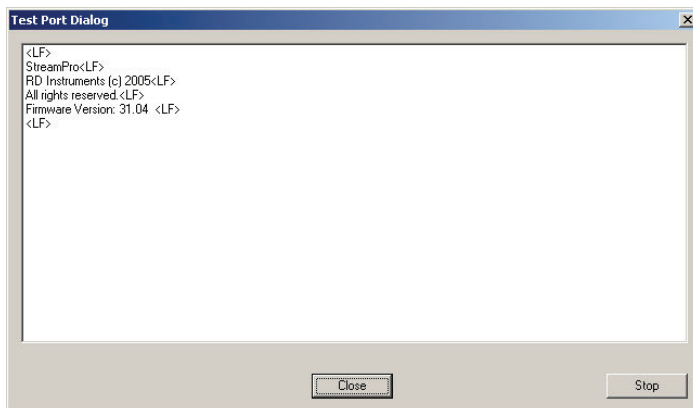


Select the communication port identified when setting up the Bluetooth device (in this example, Com Port 18).

Set the **Baudrate** to 115200.

Leave the **Databits** set to 8, **Parity** to None, and **Stopbits** set to 1.

Click **OK** to return to the **Peripheral Configuration Dialog**.



Click the **Test Port** button to connect to the StreamPro ADCP and confirm the communication setting.

You should see the wakeup message appear on the log file window.

Click the **Close** button to return to the **Peripheral Configuration Dialog**.

Click **Close** once more to exit the communication setting.



If no wakeup message appears, use *BBTalk* to troubleshoot the communications.

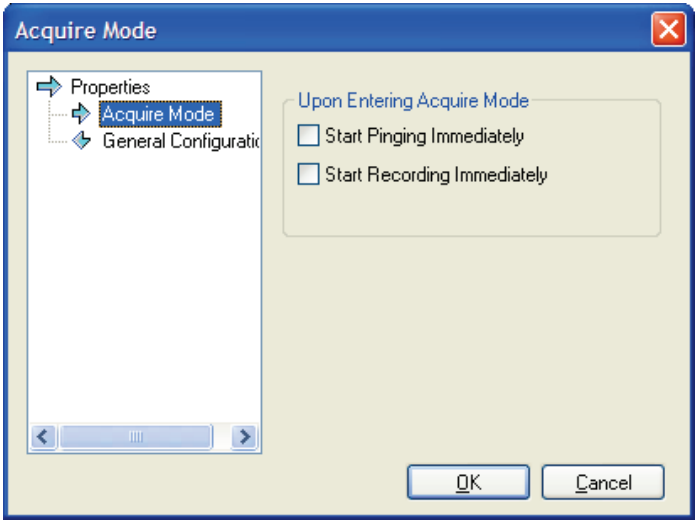
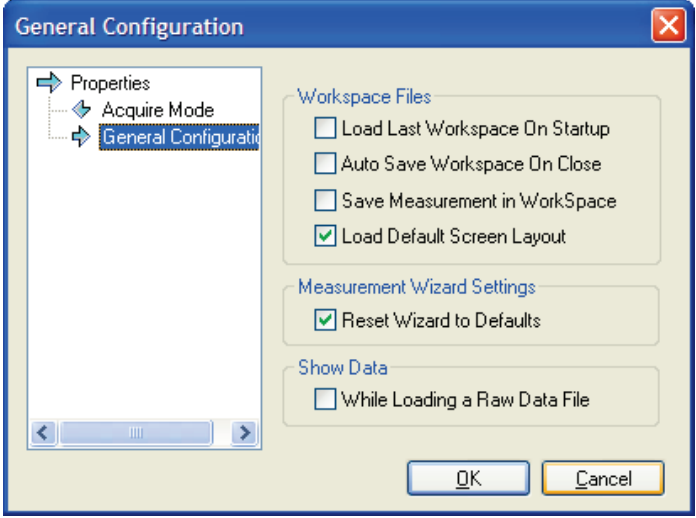


NOTE. The blue LED on the StreamPro ADCP lights when communicating with *WinRiver II SxS*. It may go off when no serial communications are in process. This is normal and does not signify any problem.

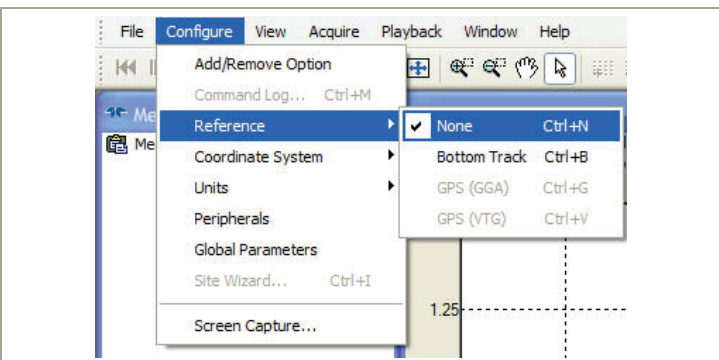
Customizing WinRiver II Section-By-Section

WinRiver II SxS can be customized to look and act as you prefer. Once setup, WinRiver II SxS will remember the settings and use them each time the program is started.

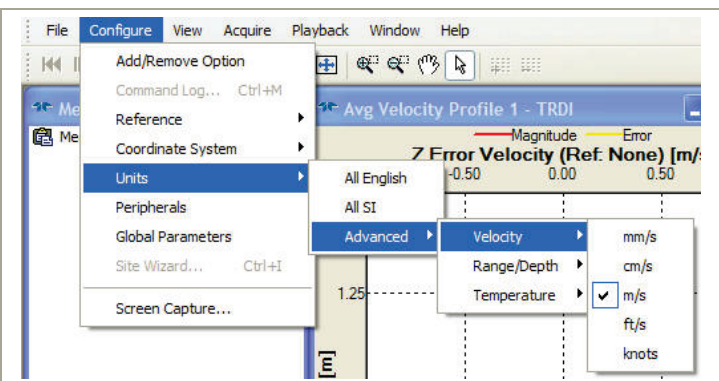
User Options

	On the File menu, click Properties. The Properties dialog sets how WinRiver II SxS behaves upon entering the Acquire mode.
	Click on General Configuration to change how workspace files are loaded or saved. Check the Reset Wizard to Defaults box to have the measurement wizard use the default settings based on a WorkHorse Rio Grande 600 kHz ADCP. Check the While Loading a Raw Data File box to begin displaying data while the file is loading. For smaller data files, this may not be noticeable.

General Preferences



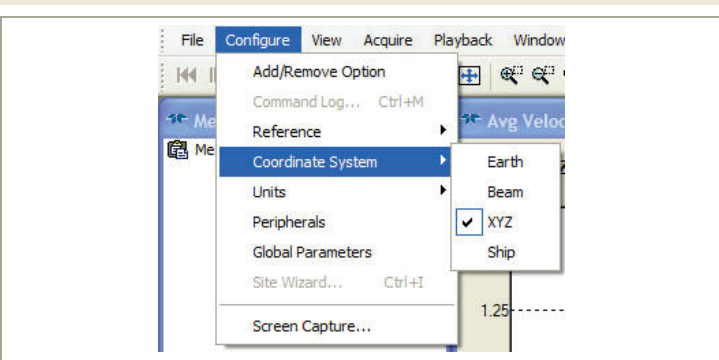
On the **Configure** menu, click **Reference**. Select the desired reference: **None** or **Bottom Track**.



To change the units for all displays, on the **Configure** menu, select **Units**.

You can change units to **All English**, **All SI** or use the **Advanced** menu and select each unit for **Velocity**, **Range/Depth**, and **Temperature**.

 When using a StreamPro ADCP on small streams/channels, change the Velocity units to mm/s or cm/sec. This will change the discharge to mm³/s or cm³/s.



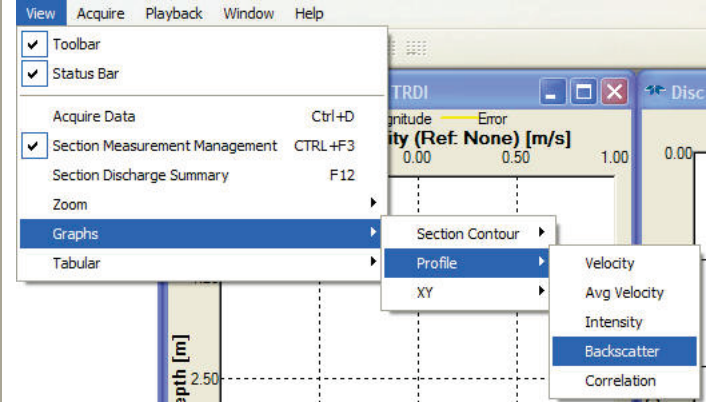
On the **Configure** menu, click **Coordinate System**.

Select the Coordinate System: **Earth**, **Beam**, **XYZ**, or **Ship**.

 TRDI recommends using **XYZ** for StreamPro ADCPs and **Earth** for Rio Grande ADCPs.

Creating Workspaces

A Workspace is a collection of windows arranged and sized as you prefer, and then saved for future use as needed. It is possible to define as many different workspaces as you would like.

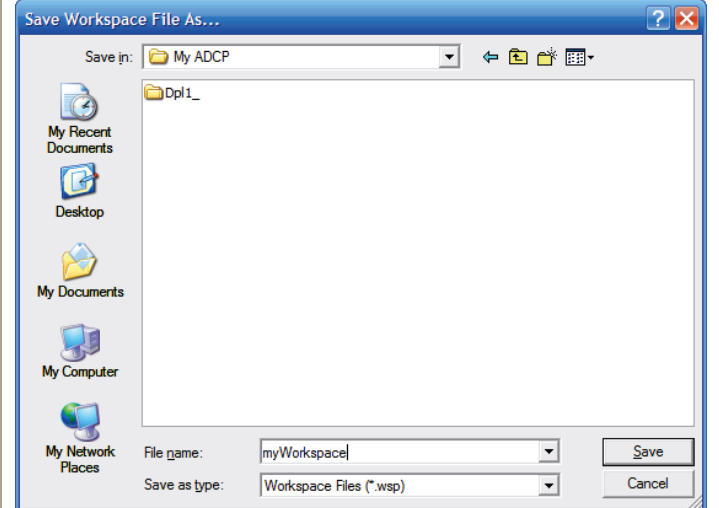


The screenshot shows the 'View' menu in the WinRiver II software. The 'Graphs' option is selected, which has opened a submenu. In this submenu, 'Section Contour' is selected, which has opened another submenu. This third submenu contains the following options: 'Velocity', 'Avg Velocity', 'Intensity', 'Backscatter', and 'Correlation'. The 'Backscatter' option is highlighted. The background shows a portion of the software interface with a graph titled 'TRDI' and a plot of 'Intensity (Ref: None) [m/s]'.

To create a Workspace file, use the **View** menu to open all the windows you want to see during data collection or playback.

Arrange the views you are interested in.

 How workspaces are loaded and saved when starting/closing *WinRiver II* SxS depends on the **User Options** (see [User Options](#)).



The screenshot shows the 'Save Workspace File As...' dialog box. The 'Save in' field is set to 'My ADCP'. The file list shows a folder named 'Dpl1_'. The 'File name' field contains 'myWorkspace' and the 'Save as type' is set to 'Workspace Files (*.wsp)'. The 'Save' button is highlighted.

When you have the displays set up the way you prefer, on the **File** menu, click **Save Workspace File As**.

To use a workspace, on the **File** menu, click **Load Workspace**.

Collecting River Discharge Data

This section has simple instructions for a typical discharge measurement.

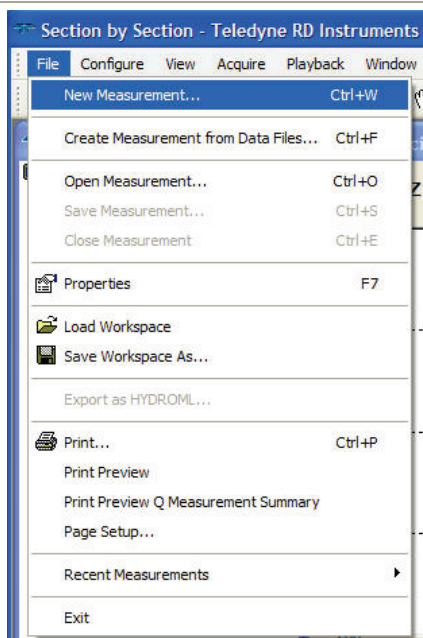
Connect the ADCP



Connect the ADCP and computer as shown in your ADCP User's Guide.

Mount the ADCP on the boat/float at the desired depth (see the ADCP Operation Manual for details).

Run the Measurement Wizard



Start *WinRiver II SxS*.

On the **File** menu, click **New Measurement** to start the Measurement Wizard.

The Measurement Wizard will create a configuration node and allow you to enter the information needed for taking measurements.

Site Information

Enter the Site Information.

Enter a **Station Name** and **Measurement Number** (alphanumeric). This can be added to the file name (see [Output Filename Options](#)).

Click **Next**.

This information will be included in the Q Measurement Summary (see [Using the WinRiver II SxS Q Measurement Summary](#)).

You can add/edit this information once the measurement wizard is completed by right-clicking on **Site Information** in the Measurement Control window and selecting **Site Wizard**.

Back **Next** **Cancel**

Click **Next** to go to the next page.

If you see this screen and the ADCP is **NOT** attached to a serial port, click **Cancel** to continue the Measurement Wizard.

If the ADCP is attached to a serial port, click **Yes** to configure the port.

The ADCP does not need to be connected to use the wizard.

If you see the following message box, this means *WinRiver II SxS* is not sure what type ADCP you are using.

Click **OK** to continue.

Configuration Dialog

Enter your choices for how the configuration will be setup.



See the following tables for details.



A green circle means that the communication port has been configured and tested. A red circle mean the ADCP has not been configured.

Devices

Enter your choices for the **Devices** section. See [Section-By-Section Communications Setup](#) for instructions on how to setup communications between *WinRiver II* SxS and the ADCP.

If you are using a Rio Grande ADCP, *WinRiver II* SxS will automatically detect and enter the **ADCP Serial Number** (the ADCP type and serial number will be entered and then grayed out).

If you are using an ADCP other than a Rio Grande, select the ADCP type from the list and enter the serial number.

Use the **Check ADCP** button to verify the ADCP communications settings.

ADCP Wizard Configuration

Max. Water Depth [m]:	5.00
Secondary Depth [m]:	0.00
Max. Water Speed [m/s]:	0.50
Streambed:	Sand
Bottom Mode:	Auto
Water Mode:	Auto
Update Rate	Auto



Leave the **Bottom** and **Water Mode** set to **Auto**. *WinRiver II SxS* will select a best suited mode based on the water depth, speed, and streambed material.

ADCP Wizard Configuration

Enter your choices for the **ADCP Configuration** section. Based on the entered information, the wizard will generate the ADCP commands.

The **Max Water Depth** and **Max Water Speed** should be defined as close as possible to the actual conditions.

Set the **Secondary Depth** to the minimum depth that will be measured in the river. Usually this will be a smaller value than the **Max Depth** or left at zero.

Streambed material should then be selected: this does not effect the configuration but it assists in producing relevant warnings.

Bottom Mode and **Water Mode** - See Table 1 for setup table.

Sample Time

Sample Time For Each Section [sec]:	40
-------------------------------------	---------------------------------

The default **Sample Time** is 40 seconds. You may increase this if desired. In slow, shallow flow increasing this value will reduce the uncertainty of the measured velocity.



You should always use a minimum Sample Time of 40 seconds.

Measurement Angle

Angle of the measurement relative to the flow [deg]	0.0
---	----------------------------------

If the measurement line is not perpendicular to the flow, then enter the angle of deviation from perpendicular in the **Measurement Angle** box. This will reduce the discharge to compensate for the fact that each panel is wider than would be the case if the measurement line were perpendicular.

Flow Type

☐ Normal (power top)

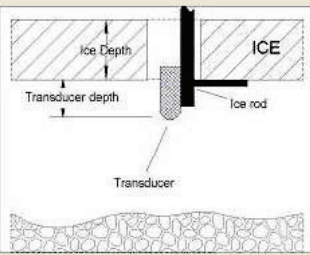
☐ Bidirectional (3-pt. slope top)

☒ Ice (no slip top)

☐ Wind driven (constant top)

Offsets

Transducer Depth [m]:



Normal is selected by default for the **Flow Type**.

If the velocity profile is expected to be non-classical, wind driven at the surface for example, then selecting the **Bidirectional** box will use 3-point extrapolation at the surface and no-slip extrapolation to the bottom.

If you are measuring below ice cover, check the **Ice** box and enter the **Transducer Depth** from bottom of the ice cover to the center of the ADCP transducer faces.

Offsets

Transducer Depth [m]:

Use the **Transducer Depth** field to set the depth from the water surface to the center of the ADCP transducer faces.



The transducer depth should be checked periodically during data collection.

 **Back** **Next**   **Cancel**

Click **Next** to go to the next page.

Output Filename Options

Setup Dialog

Measurement Wizard

Output Filename Options

Recording

Filename Prefix: ☒ Use Prefix in Filename

Output Directory:

Include Filename Options

☒ Measurement Number

☐ Sequence Number - Maximum File Size (Mb):

Use Date/Time in Filename

☒ No Date/Time

☐ Short (YY-MM-DD)

☐ Long (YY-MM-DD hhmmss)

Use Delimiter in Filename

☐ No Delimiter

☒ Underscore

☐ Custom

Filename Preview

C:\Measurements\Station\Station_0.PD0

Teledyne RD Instruments

If a **Station Name** was entered on the first page of the wizard, it will be used in the **Filename Prefix** box.

WinRiver II SxS uses the **Filename Prefix** to create the data file names made during data collection.

Use the **Output Directory** field to select where the data file will be stored.

Check the **Measurement Number** box to add it to the file name (see [Site Information](#)).

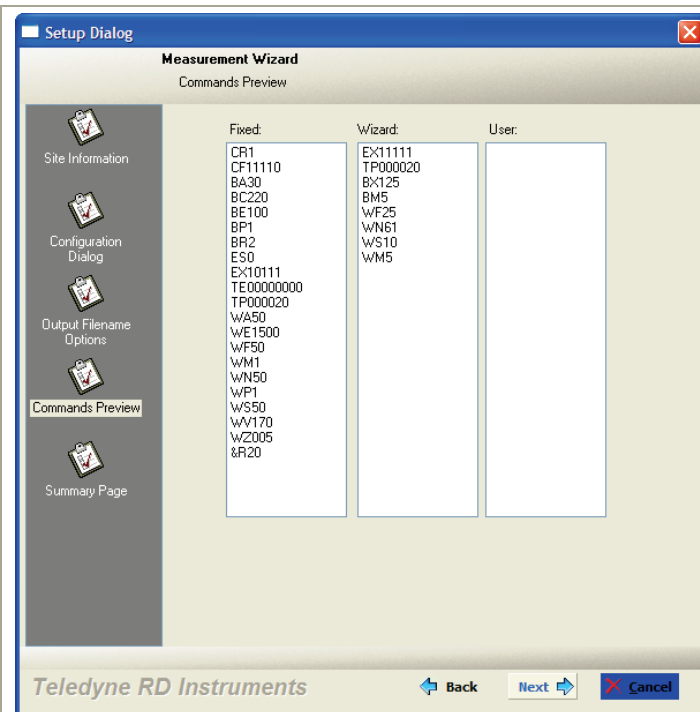
Check the **Sequence Number** box and enter a **Max File Size** if you want to limit the size of the data file. Once the file size has been reached, the sequence number will increment.

To add the Date/Time to the filename, check the **Short (YY-MM-DD)** or **Long (YY-MM-DD hhmmss)** button.

Select what type delimiter to use in the filename by selecting **No Delimiter**, **Underscore**, or **Custom**.

Click **Next** to go to the next page.

Commands Preview



See **Error! AutoText entry not defined.2, Error! AutoText entry not defined.43** for a short list of ADCP commands commonly sent via *WinRiver II SxS*.

The Commands Preview is where adjustments can be made to the commands produced by the measurement wizard.

There are three columns - the first is the **Fixed** commands that are sent by default by *WinRiver II SxS* to the ADCP.

The second column is commands that are produced by the **Wizard**; these commands will overwrite the corresponding original fixed commands.

The final column is for **User** commands; these can be added to by the user, and just as before, these commands will overwrite the previous column.

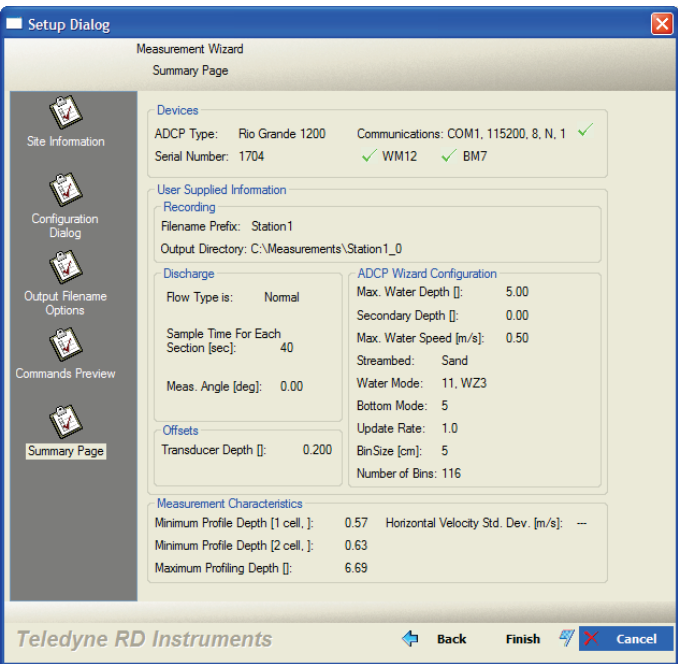


Do not enter any commands in the User section unless you are fully aware of what the command does.

Back Next Cancel

Click **Next** to go to the next page.

Summary Page



Review the **Summary Page**.

 A green check mark next to **WM12** and **BM7** means that the ADCP is capable of using these modes, not that the mode is selected.

View the **ADCP Wizard Configuration** section to see what water and bottom mode is selected.

 Any calculations displayed on the Summary page do **not** take into account any **User** commands (see [Commands Preview](#)).

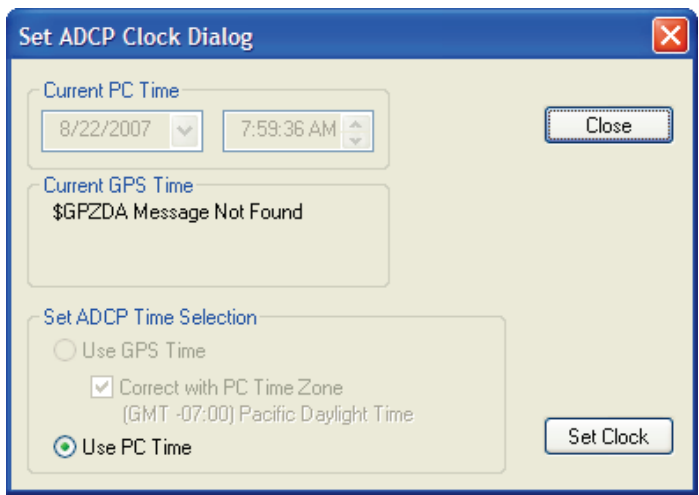


When done, click **Finish**.

QA/QC Items

Before taking measurements, check the following items.

Set ADCP Clock



The dialog box titled "Set ADCP Clock Dialog" has a blue header bar with a close button (X). It contains three sections: "Current PC Time" with a date dropdown (8/22/2007) and a time spinner (7:59:36 AM); "Current GPS Time" showing "\$GPZDA Message Not Found"; and "Set ADCP Time Selection" with radio buttons for "Use GPS Time" and "Use PC Time", and a checked checkbox for "Correct with PC Time Zone (GMT -07:00) Pacific Daylight Time". There are "Close" and "Set Clock" buttons.

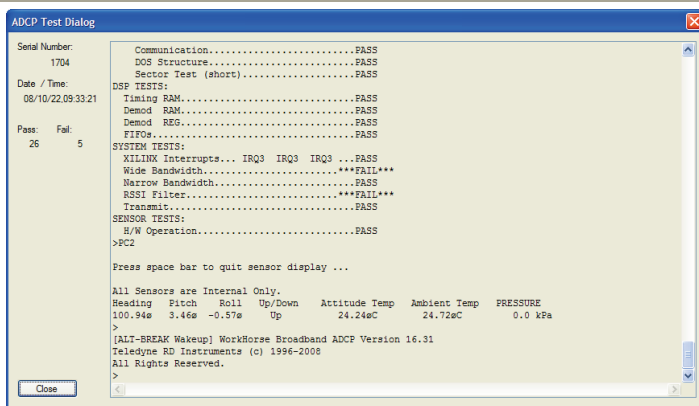
On the **Acquire** menu, click **Set ADCP Clock**.

Click the **Set Clock** button to set the ADCP's time to the GPS time (if available) or the PC's time. If necessary, set the PC's clock first.

 The first time a transect is started, the **Set ADCP Clock** dialog will open.

 StreamPro ADCPs users will notice that the blue LED will flash until the **Close** button is clicked. This is normal. The TS command is send once per second until the close button is clicked.

Test ADCP



The dialog box titled "ADCP Test Dialog" has a blue header bar with a close button (X). It displays test results for Serial Number 1704, Date/Time 08/10/22 09:33:21, and Pass/Fail counts (26/5). The tests include Communication, DOS Structure, Sector Test, DSP TESTS (Timing RAM, Demod RAM, Demod REG, FIFOs), SYSTEM TESTS (XILINK Interrupts, Wide Bandwidth, Narrow Bandwidth, RSSI Filter, Transmit), and SENSOR TESTS (H/W Operation). A table shows sensor data: Heading (100.94e), Pitch (3.46e), Roll (-0.57e), Up/Down (Up), Attitude Temp (24.24eC), Ambient Temp (24.72eC), and PRESSURE (0.0 kPa). The footer includes "[ALT-BREAK Wakeup] WorkHorse Broadband ADCP Version 16.31" and "Teledyne RD Instruments (c) 1996-2008". There is a "Close" button at the bottom left.

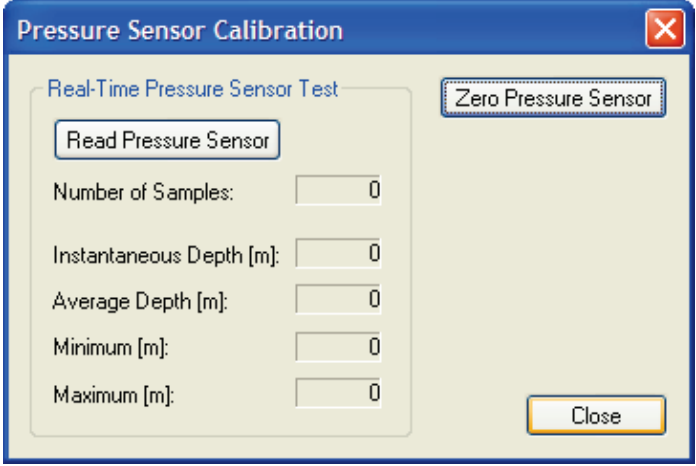
On the **Acquire** menu, click **Execute Test ADCP** to verify the ADCP is functioning properly.

Click the **Stop PC2** button to end the PC2 test. Click **Close** to exit the **Test ADCP** dialog.

 The tests should be run while the ADCP is in non moving water. Running the test in air will not harm the ADCP, but some tests may fail in air.

 Do not use the **RGTest** on a StreamPro ADCP. Use **BBTalk** to test a StreamPro ADCP.

Test Pressure Sensor



On the Acquire menu, click **Execute Pressure Sensor Test** to verify the ADCP's pressure sensor is functioning properly.

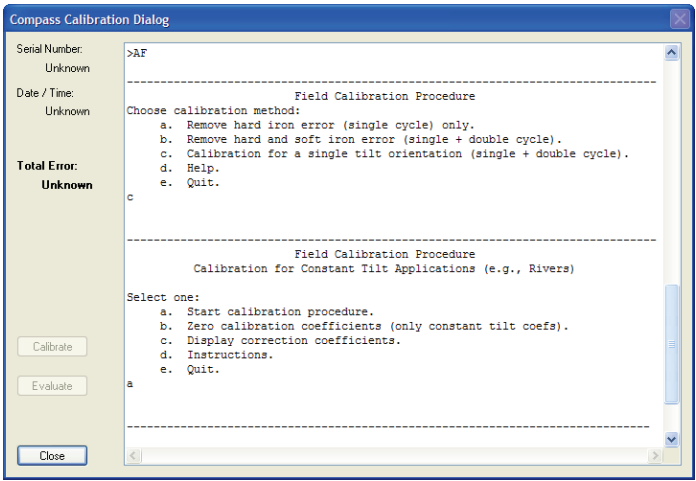
Click the **Read Pressure Sensor** button to get samples.

Click the **Zero Pressure Sensor** button to zero out the sensor.

 If your ADCP does not have a pressure sensor, you will see an error message.

 Do not use the **Pressure Sensor Test** on a StreamPro or RiverRay ADCP. StreamPro and RiverRay systems do not have a pressure sensor.

Compass Calibration



On the **Acquire** menu, click **Execute Compass Calibration**.

Click on the **Calibrate** button to begin the compass calibration.

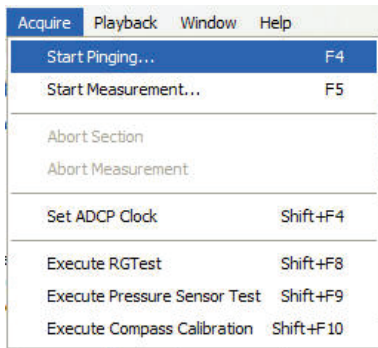
 For more information on the Compass Calibration, see the WinRiver II User's Guide, Integrating Depth Sounder, External Heading, and GPS Data.

Measurements

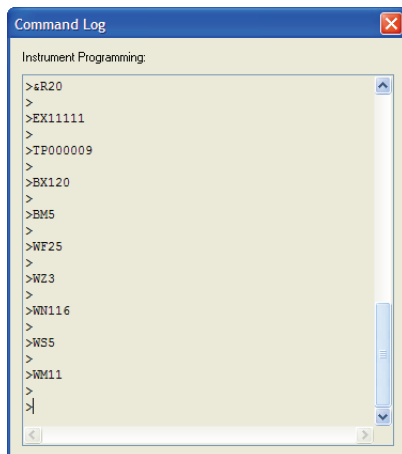
Best practices encourage collecting 20 or more sections. Each section should be sized so that it has less than 5% of the total rated discharge Q . The following sequence must be repeated for each measurement.



NOTE. TRDI suggests doing a quick survey of the stream using the *WinRiver II* software to get a value for the rated Q , if it is not known for the site.



Click the **Acquire** menu and select **Start Pinging** to get velocity data and depth from the ADCP.



SxS will send the commands generated by the configuration wizard to the ADCP.

Review the **Command Log** to see if any error occurs.



If the command log closed before you finished reviewing the commands, click **Configure, Command Log...** Scroll through the log file and make sure no command generated an error message.

Composite Tabular 1 - TRDI			
Current Section # 21			
Nmb. of Ens.	Lost Ens.	Bad Ens.	
40	0	2	
Ref: None			
Ensemble Data			
January 07, 2005 13:22:56.50			
Ens. #	Good Bins	Delta Time	
40	4	1.00	
Pitch	Roll	Heading	
0.00°	0.00°	0.00°	
Temp.	Press. Sensor		
14.81°C	NA		
Water Speed	0.380	[m/s]	
Calc. Depth	0.64	[m]	
Section Data			
Duration	39.00	[s]	
Mean Speed	0.281	[m/s]	
CV Wtr Spd	20.73	[%]	
Std. Flow Dir.	12.63	[°]	
Mean Depth	0.62	[m]	
Top Q	0.026	[m ² /s]	
Measured Q	0.051	[m ² /s]	
Bottom Q	0.030	[m ² /s]	
Total Q	0.108	[m ² /s]	
Accumulated Section Data			
Total Q	1.938	[m ² /s]	
Total Area	7.604	[m ²]	
Mean Speed	0.255	[m/s]	

If everything is working properly, you will see the **Water Speed** in the profile, the **Calculated Depth** of the water, the **Coefficient of Variation** of the velocity measurement, and the **Number of Good Bins** at that location.

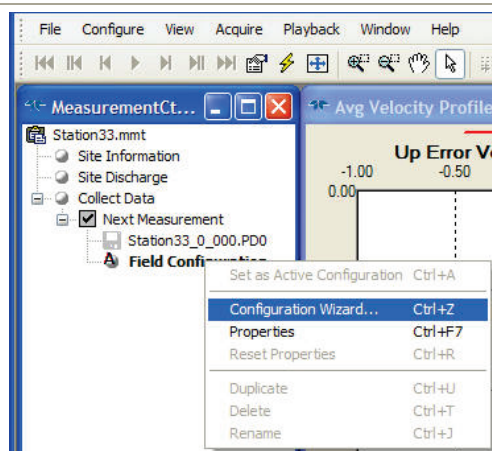
The **Coefficient of Variation** value should slowly decrease as more pings are averaged together. This will give you a good idea of how long a period is required to obtain the desired stability in the velocity measurement. You can use this to determine whether to increase or decrease the measurement time in the configuration file.

You can also make a quick crossing to make sure that your maximum depth value is correct and to get an idea of where you can start and stop measuring at each side.



NOTE. Coefficient of Variation for *Section-By-Section* is defined as (Standard Deviation of the average velocity for each ping)/(average velocity of all pings).

Where the average velocity is the average of all good cells in the profile.



If the field configuration needs to be adjusted, on the **Acquire** menu, click **Stop Pinging**.

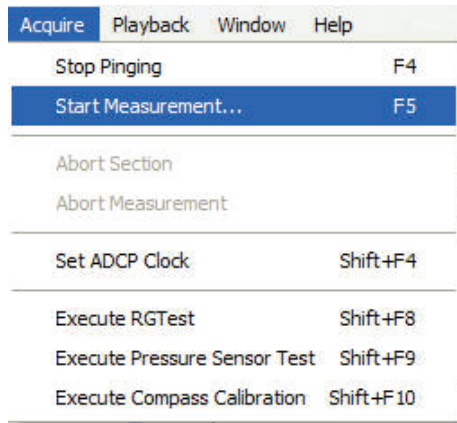
Right-click on the **Field Configuration** node and select **Configuration Wizard**. This will restart the Measurement Wizard (see [Configuration Dialog](#)).

Make your adjustments.

To restart, click the **Acquire** menu and select **Start Pinging**.



Once a measurement has started, the field configuration node can no longer be changed.



Move the ADCP to the start point of the measurement.

Click the **Acquire** menu and select **Start Measurement**.

Since this is the first measurement, it will be at an edge. Select the edge you will start from (**Left Edge** or **Right Edge**) and enter the position of the edge in the **Edge of Water** box.

If the edge is vertical instead of sloped, enter the water depth at the edge and enter the correction factor. This will be applied to the velocity measured at the next section and used to estimate the velocity in section 1. The correction factor is usually equal to 1.00.

Click **OK** to continue.

Now move to the first vertical to be measured.

The **Section Flow Angle** setting is defaulted to the **Measurement Angle** entered in the Site Configuration Wizard. If needed, enter the **Section Flow Angle** for the current station. The **Correction Factor** is the cosine of the **Flow Angle**.

Enter the position of the **Current Station**.

Check the **Flow Type**.

Click **OK** to continue.

Composite Tabular 1 - TRDI			
Current Section # 2			
Nmb. of Ens.	Lost Ens.	Bad Ens.	
40	0	2	
Ref: None			
Ensemble Data			
January 07, 2005 13:22:56.50			
Ens. #	Good Bins	Delta Time	
40	4	1.00	
Pitch	Roll	Heading	
0.00°	0.00°	0.00°	
Temp.	Press. Sensor		
14.81°C	NA		
Water Speed	0.380	[m/s]	
Calc. Depth	0.64	[m]	
Section Data			
Duration	39.00	[s]	
Mean Speed	0.281	[m/s]	
CV Wtr Spd	20.73	[%]	

Now the data will be recorded.

The **Duration** near the center of the **Composite Tabular 1** screen will count up to the time remaining in the measurement interval (default 40 seconds).

As the ADCP pings, the **Average Velocity** of all of the pings is displayed as well as the **Coefficient of Variation** of all of the pings to that time. The Coefficient of Variation should slowly decrease and reach a relatively stable value. The depth of the water is also displayed along with the number of seconds that have elapsed since pinging began.



NOTE. If you want to stop the measurement at any time, select **Acquire, Abort Section**. This may be because you realized that you forgot to move to the next station. A message will appear asking if you are sure you want to stop taking data. Answer **Yes** to stop and move on.

Post Measurement Dialog	
Recorded Data for Measurement: 3	
Calculated Discharge [m³/s]:	0.096
Calculated Discharge % Rated:	0.0
Average Velocity [m/s]:	0.536
Coefficient of Variation [%]:	12.3
Average Depth [m]:	0.718
Number of Valid Pings:	40
Section Flow Angle CF:	1.00
Standard Deviation Flow Direction:	6.33
Past Discharge Data	
Previous Section Discharge [m³/s]:	0.071
Accumulated Section Discharge [m³/s]:	0.167
OK Cancel	

When the measurement interval has expired, this page will appear giving a summary of the data for that station. The discharge calculated for that panel is shown followed by the percentage of the rated Q that is measured at this station.

If the discharge for the station is more than 5% of the rated Q, these values will be shown in red text; otherwise, they will be shown in black text.

If you are satisfied with the results for this station, click **Accept Data**. If you want to repeat the station or if you want to change the position before repeating, click the **Reject and Repeat** button.

Pre Measurement Dialog

Section Number: 3 ☐ Edge

Acquisition interval for Section [sec]: 40

Section Flow Angle [deg]: 0.00

Correction Factor: 1.00

Please, Enter distance to IP

Previous Station [m]: 0.25

Current Station [m]: 0.50

OK

If you accepted the previous data Press **F5** to start a new section. The **Section Number** is incremented by one, from 2 to 3.

The **Previous Station's** position and the value that was **Next Station** is now the **Current Station**.

The spacing between the previous section's **Current** and **Next Station** is added to the **Current Station** and displayed as **Next Station**. If you want to change the position of the next station, then enter that position.

Don't forget to move to the location of the **Current Station**. After checking to be sure you are at the correct position, click **OK** to begin taking data.

Post Measurement Dialog

Recorded Data for Measurement: 3

Calculated Discharge [m³/s]: **0.096**

Calculated Discharge % Rated: **0.0**

Average Velocity [m/s]: **0.536**

Coefficient of Variation [%]: **12.3**

Average Depth [m]: **0.718**

Number of Valid Pings: **40**

Section Flow Angle CF: **1.00**

Standard Deviation Flow Direction: **6.33**

Past Discharge Data

Previous Section Discharge [m³/s]: **0.071**

Accumulated Section Discharge [m³/s]: **0.167**

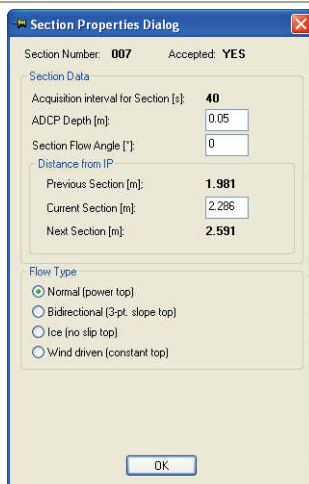
OK Cancel

As each station is completed, click the **Accept Data** or **Reject and Repeat** button.

Composite Tabular 1 - TRDI

Sect#	Ens#	Good Bins	Delta Time
3, 798	3		0.20
November 20, 2007 10:16:03.93			
Pitch	Roll	Heading	
3.96°	-1.24°	86.31°	
Temp.	Press Sensor		
26.65°C	NA		
Velocity (Ref: None)			
Water Speed	0.575	[m/s]	
Calc. Depth	0.55	[m]	
Section Data			
Nmb. of Ens.	Lost Ens.	Bad Ens.	
200	0	2	
Good Bins	19		
Coeff. Var	137.84	[%]	
Duration	40.61	[s]	
Discharge (Ref: None) Right to Left			
Top Q	0.015	[m ³ /s]	
Measured Q	0.111	[m ³ /s]	
Bottom Q	0.190	[m ³ /s]	
Total Q	0.315	[m ³ /s]	
Accumulated Section Data			
Discharge (Ref: None) Right to Left			
Total Q	0.622	[m ³ /s]	
Total Area	1.221	[m ²]	
Avg. Velocity	0.509	[m/s]	

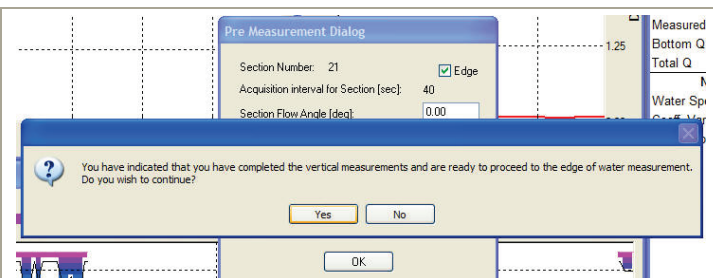
The Composite Tabular screen will display numbers in blue when SxS is waiting for the next section to be started.



The 'Section Properties Dialog' box is shown. It has a title bar with a close button. The 'Section Number' is 007 and 'Accepted' is YES. Under 'Section Data', 'Acquisition interval for Section [s]' is 40, 'ADCP Depth [m]' is 0.05, and 'Section Flow Angle [°]' is 0. Under 'Distance from IP', 'Previous Section [m]' is 1.981, 'Current Section [m]' is 2.286, and 'Next Section [m]' is 2.591. Under 'Flow Type', 'Normal (power top)' is selected with a radio button. Other options are 'Bidirectional (3-pt. slope top)', 'Ice (no slip top)', and 'Wind driven (constant top)'. An 'OK' button is at the bottom.

Continue taking data at successive stations.

Remember to move to the next station and press **F5** to start.

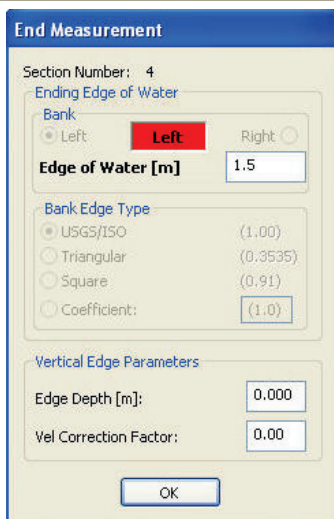


The 'Pre Measurement Dialog' box is shown. 'Section Number' is 21, 'Acquisition interval for Section [sec]' is 40, and 'Section Flow Angle [deg]' is 0.00. The 'Edge' checkbox is checked. Below it, a message box says: 'You have indicated that you have completed the vertical measurements and are ready to proceed to the edge of water measurement. Do you wish to continue?' with 'Yes' and 'No' buttons. An 'OK' button is at the bottom of the dialog.

Check the **Edge** box when you reach the last measurement position.

As soon as the **Edge** box is checked, then this screen will appear.

Click **Yes** to finish the measurement.



The 'End Measurement' dialog box is shown. 'Section Number' is 4. Under 'Ending Edge of Water', 'Bank' has 'Left' selected with a radio button and highlighted in red. 'Edge of Water [m]' is 1.5. Under 'Bank Edge Type', 'USGS/ISO' is selected with a radio button, with values (1.00), (0.3535), (0.91), and (1.0) listed. Under 'Vertical Edge Parameters', 'Edge Depth [m]' is 0.000 and 'Vel Correction Factor' is 0.00. An 'OK' button is at the bottom.

The selected edge will be the opposite edge from the one you selected at the beginning.

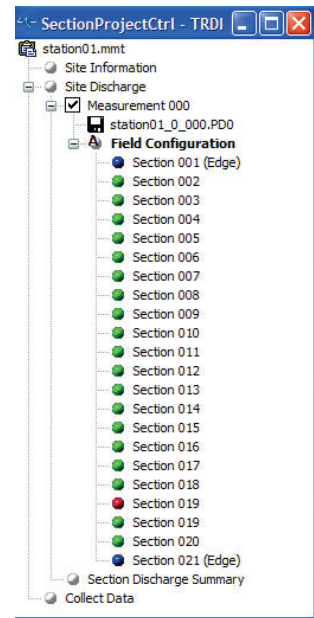
Enter the position of the final edge.

If this edge is a vertical wall, then also enter the depth and the appropriate correction factor.

Click **OK**.

Step by Step Data Collection

1. Open or create a measurement file.
2. Press **F4** to start ping.
3. At Section 1 (Edge), press **F5** to start the measurement.
4. Enter the starting distance from the shore.
5. Select **Left** or **Right** bank.
6. Move to Section 2.
7. Press **F5** to start section measurement. Click **OK**.
8. Wait 40 seconds while data is recorded.
9. Accept/Reject data.
10. Move to next section.
11. Repeat steps 7 through 10 to collect at least 20 sections that agree with each other within 5% of the mean of all the samples.
12. At the other bank of the river, select **Edge**.
13. Enter the ending distance from the shore.



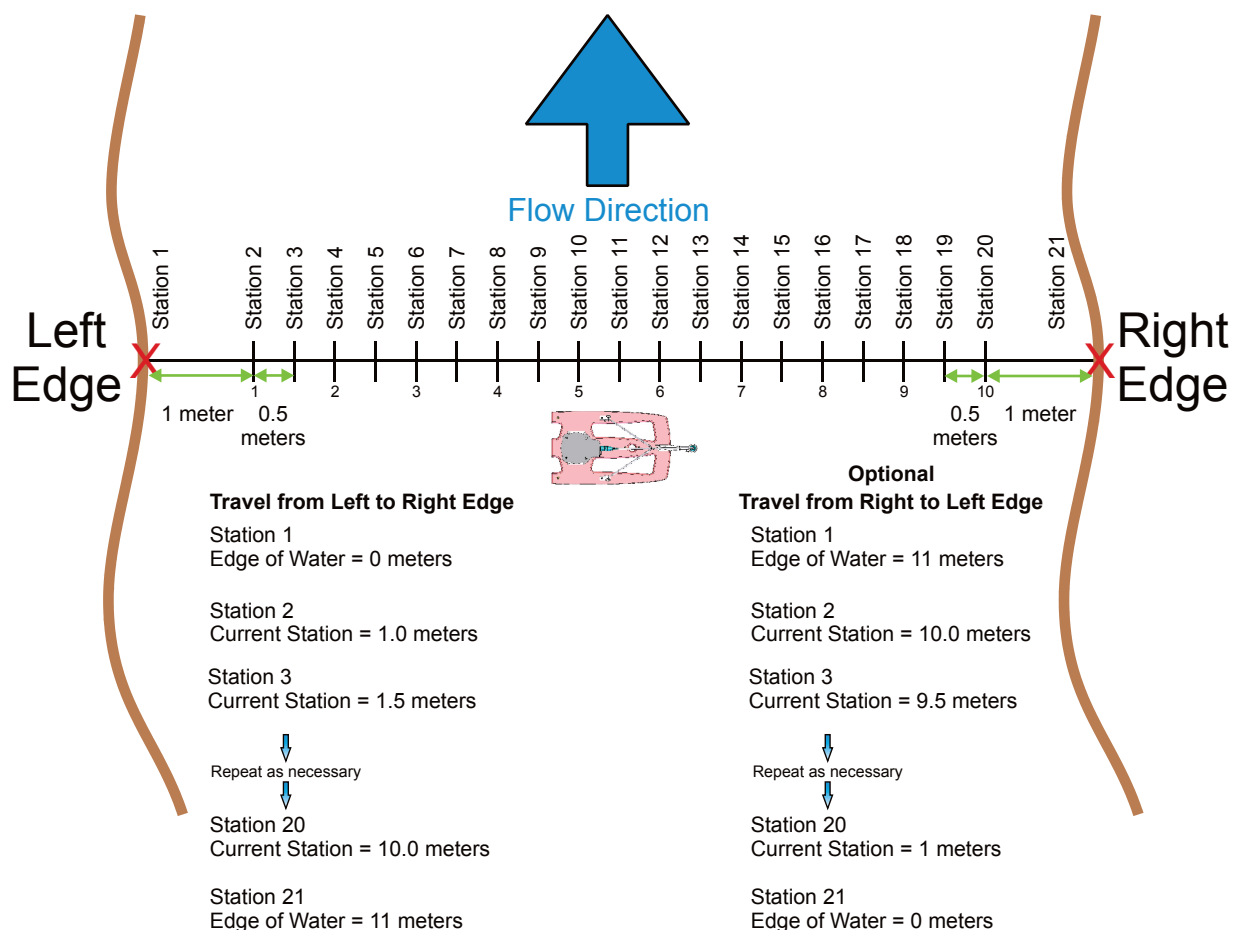


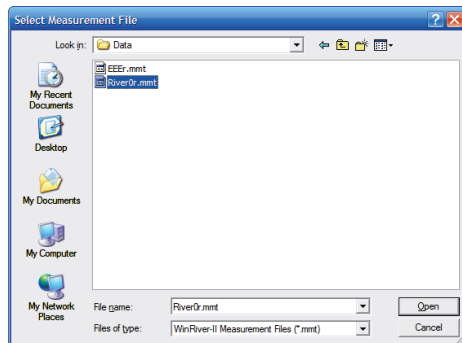
Figure 1. Overview of Data Collection

Data Collections Tips

- Check the **Edge** box (**Left** or **Right Edge**) for the first and last sections and enter the distance of the edge of the water. If the edge is vertical instead of sloped, enter the **Edge Depth** and enter the **Correction** factor.
- Minimize the ADCP movement while section measurements are recorded.
- Collect 20 or more sections.
- TRDI suggests doing a quick survey of the stream using the *WinRiver II* software to get a value for the rated Q , if it is not known for the site.
- Each section should be sized so that it has less than 5% of the total rated discharge Q .
- Station distances can count up (1.5 meters, next station 2.0 meters) or down (2.0 meters, next station 1.5 meters) as necessary.

Viewing Data with *WinRiver II SxS*

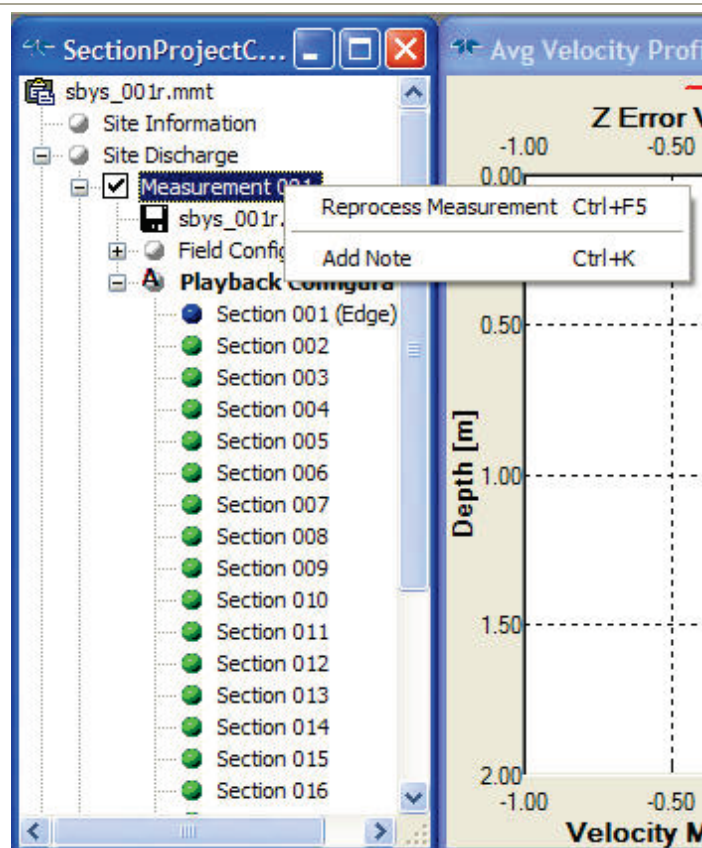
WinRiver II SxS is used for post-processing data to get a total discharge value for the channel. After collecting 20 sections on the water body, each station must be verified to be within 5% of the mean discharge calculated for the set. If any of the stations are outside of the tolerance, additional measurements should be taken.



Start *WinRiver II SxS*.

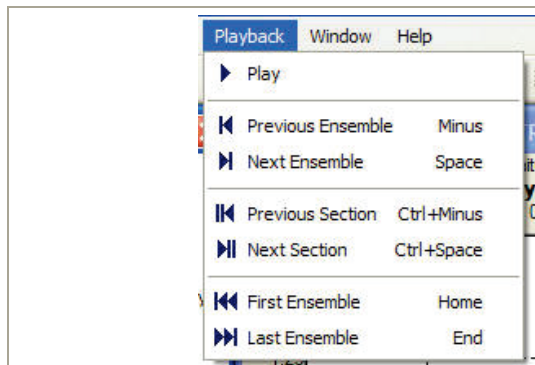
On the **File** menu click **Open Measurement...**

Select the measurement file (*.mmt) to be played and click **Open**.



To playback a measurement file, right-click on the measurement file and select **Reprocess Measurement**. This will create a copy of the **Field Configuration** node and creates a **Playback Configuration** node.

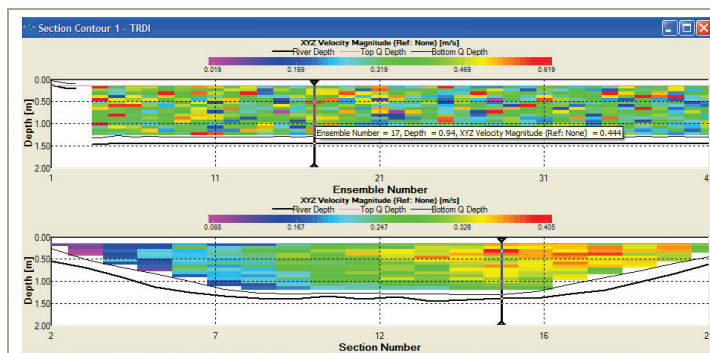
The data files will automatically play to the end of the file.



On the **Playback** menu, click **First Ensemble** to go to the beginning of the measurement file. Click **Play** to review the data.



The playback tool bar also has functions to start, stop, rewind, and go to the end of the data file.



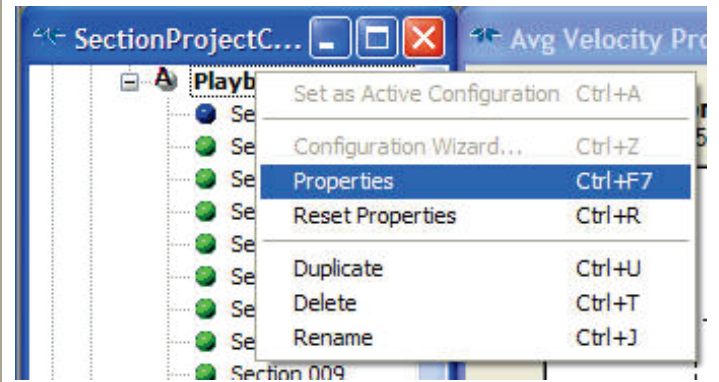
To quickly play through the data, on the **Playback** menu, drag the ensemble marker on the top contour plot.

Drag the section marker on the bottom contour plot to quickly select a Section Number.

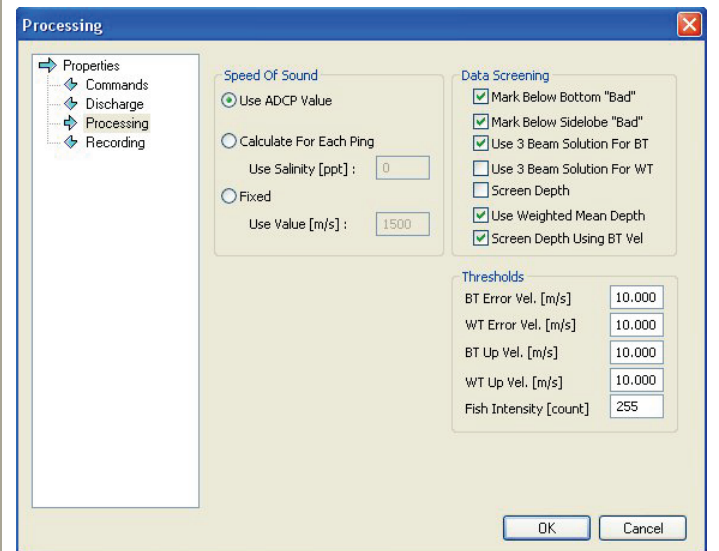
Reprocessing Data

WinRiver II SxS is used for post-processing data to get a total discharge value for the channel. Common post-processing tasks include exporting data for use in other programs and make corrections to the configuration nodes.

Corrections to the Playback Configuration Node



If a property needs correction, right-click on the **Playback** configuration node and select **Properties**.



Check **Processing**.

Check the post-processing options.

Click **OK**.

 These settings apply to all sections.

Change Section Angle

Section Properties Dialog

Section Number: **002** Accepted: **YES**

Section Data

Acquisition interval for Section [s]: **40**

ADCP Depth [m]:

Section Flow Angle [°]:

Distance from IP

Previous Section [m]: **8.300**

Current Section [m]:

Next Section [m]: **7.000**

Flow Type

☒ Normal (power top)

☐ Bidirectional (3-pt. slope top)

☐ Ice (no slip top)

☐ Wind driven (constant top)

OK

On the **Measurement Control** window, right-click on a **section** node and select **Properties**.

If data was collected using an incorrect ADCP depth, enter the correct ADCP depth.

If the measurement line was not perpendicular to the flow, then enter the angle of deviation from perpendicular in the **Section Flow Angle** box.

You can also correct the **Current Section** distance and the **Flow Type**.



These corrections need to be done for each section.

Click **OK**.

Change Edge Shape

Section Properties Dialog

Section Number: **Edge** Accepted: **YES**

Section Data

Acquisition interval for Section [s]: **40**

ADCP Depth [m]:

Section Flow Angle [°]:

Distance from IP

Previous Section [m]: **N/A**

Current Section [m]:

Next Section [m]: **7.300**

Edge Data

Left/Right Edge: **Left**

Edge Depth [m]:

Correction Factor:

Edge Shape:

☒ USGS/ISO (1.00)

☐ Triangular (0.3535)

☐ Square (0.91)

☐ User:

OK

On the **Measurement Control** window, right-click on the **edge** node and select **Properties**.

Select the **Edge Shape**.

Section Discharge Summary

Measurement	Q [m³/s]	% Total Q	Accum Q [m³/s]	% Diff. Rating	% Rated Q	Avg Velocity [m/s]	Coef. Var [%]	Width [m]	Distance [m]	Depth [m]	Area [m²]	Flow Angle Coeff.
Measurement ...	3.149											
Section 001	0.000	0.00%	0.000		0.00%	0.000	0.00%	0.000	0.000	0.000	0.000	1.000
Section 002	0.017	0.56%	0.017		3.77%	0.059	60.13	0.518	0.762	0.546	0.283	1.000
Section 003	0.022	0.70%	0.040		5.36%	0.089	49.69	0.302	1.060	0.784	0.223	1.000
Section 004	0.044	1.39%	0.084		10.11%	0.153	29.57	0.304	1.172	0.957	0.251	1.000
Section 005	0.057	1.81%	0.141		12.96%	0.164	25.36	0.304	1.676	1.159	0.353	1.000
Section 006	0.075	2.38%	0.217		17.24%	0.196	21.89	0.305	1.981	1.271	0.388	1.000
Section 007	0.080	2.54%	0.297		18.23%	0.183	20.57	0.305	2.286	1.360	0.415	1.000
Section 008	0.090	2.86%	0.387		20.46%	0.209	19.21	0.305	2.591	1.415	0.431	1.000
Section 009	0.103	3.27%	0.490		23.51%	0.239	11.96	0.304	2.896	1.423	0.433	1.000
Section 010	0.114	3.62%	0.605		25.10%	0.253	11.19	0.304	3.200	1.423	0.433	1.000
Section 011	0.130	4.13%	0.735		27.36%	0.277	13.99	0.305	3.505	1.425	0.434	1.000
Section 012	0.137	4.35%	0.872		28.91%	0.291	12.76	0.305	3.810	1.423	0.437	1.000
Section 013	0.128	4.06%	0.980		29.03%	0.290	11.96	0.305	4.115	1.444	0.440	1.000
Section 014	0.137	4.35%	1.117		31.00%	0.310	10.12	0.304	4.420	1.447	0.441	1.000
Section 015	0.144	4.57%	1.261		32.66%	0.320	10.24	0.304	4.724	1.446	0.440	1.000
Section 016	0.137	4.35%	1.398		31.10%	0.320	9.06	0.305	5.029	1.406	0.420	1.000
Section 017	0.133	4.22%	1.531		30.13%	0.323	11.48	0.305	5.334	1.351	0.400	1.000
Section 018	0.120	3.81%	1.651		27.22%	0.321	10.04	0.305	5.639	1.224	0.373	1.000
Section 019	0.102	3.24%	1.753		20.24%	0.316	15.30	0.304	5.944	1.098	0.316	1.000
Section 020	0.081	2.57%	1.834		18.34%	0.321	15.86	0.304	6.248	0.926	0.251	1.000
Section 021	0.055	1.75%	1.889		10.83%	0.277	20.98	0.303	6.553	0.620	0.218	1.000
Section 022	0.000	0.00%	1.889		0.00%	0.000	0.00%	0.457	7.468	0.000	0.000	1.000

The **Section Discharge Summary** is an easy way to review sections. It is opened by selecting **View, Section Discharge Summary** or by using the shortcut key **F12**.

The section discharge summary shows all recent sections that have been made and all relevant information on them. It can be used after data collection and during playback and is very effective at establishing whether a measurement is good or not.



As a standard for all ADCP users it is widely accepted that sections should be within 5% of each other assuming constant stage.

Export Data

Measurement	Q [m³/s]	% Total Q	Accum Q [m³/s]	% Diff.
Measurement ...	3.149			4.9
Section 001	0.000	0.00%	0.000	
Section 002	0.135	4.28%	0.135	
Section 003	0.090	2.85%	0.224	
Section 004	0.113	3.59%	0.337	
Section 005	0.142	4.50%	0.479	
Section 006	0.164	5.19%	0.643	
Section 007	0.187			
Section 008	0.194			
Section 009	0.187			
Section 010	0.182			
Section 011	0.177	5.61%	1.570	
Section 012	0.180	5.71%	1.750	
Section 013	0.180	5.72%	1.930	
Section 014	0.184	5.85%	2.114	
Section 015	0.188	5.97%	2.303	
Section 016	0.174	5.52%	2.476	
Section 017	0.171	5.44%	2.648	
Section 018	0.148	4.72%	2.796	
Section 019	0.130	4.12%	2.926	
Section 020	0.093	2.95%	3.019	
Section 021	0.077	2.44%	3.095	
Section 022	0.054	1.70%	3.149	
Section 023	0.000	0.00%	3.149	

Right-click the **Section Discharge Summary** display and select **Export to CSV** to export a comma delimited file able to be used in spreadsheets and other programs.

Select **Copy** to copy the text to the Windows® clipboard as tabular delimited text.

Use the **Properties** function to change the type font used for the text.

Using the WinRiver II SxS Q Measurement Summary

The Q Measurement Summary creates a summary of the measurement that can be printed.

Teledyne RD Instruments
Section-By-Section Discharge Summary Date: 01/07/2005

Site Information		Measurement Information	
Site Name: chk13		Units of Measure: SI	
		Measure No.: 0	

Summary:
 Width: 7.438 Area: 7.604 Velocity: 0.255 Stage: 1.000 Discharge: 1.938

Personnel & Equipment
 Party: me

Rating Information
 Est. Uncert.: 3.492 L. Edge Type: USGS/ISO R. Edge Type: USGS/ISO
 Num. Stations: 22 L. Edge Coeff.: 1.000 R. Edge Coeff.: 1.000
 Rated Q: 0.440 Begin Shore: RIGHT

System Information		System Setup	
Serial Number: 208		ADCP Depth: 0.050	Meas. Angle: 0.000
Firmware Version: 31.03		Cell Size: 0.07	Num. Cells: 20
Software Version: 2.02.18.00			
System Frequency: 2.0 MHz			

Stn. #	Distance (m)	Depth (m)	Width (m)	Start Time	Meas. Time (s)	Percent Good	Coeff. of Variation	Average Velocity (m/s)	Adjusted Velocity (m/s)	Angular Corr. Factor	Area (m ²)	Discharge (m ³ /s)	% of Total
1	0.030	0.000	0.366	12:59:05	0.00	97.44	0.000	0.000	0.000	1.000	0.000	0.000	0.00
2	0.762	0.546	0.518	12:59:05	38.00	97.44	60.126	0.059	0.059	1.000	0.283	0.016	0.81
3	1.067	0.764	0.305	12:59:43	40.00	87.80	43.693	0.099	0.099	1.000	0.233	0.023	1.19
4	1.372	0.962	0.304	13:02:04	41.00	90.48	28.743	0.154	0.154	1.000	0.293	0.045	2.32
5	1.676	1.159	0.304	13:03:20	40.00	95.12	25.359	0.161	0.161	1.000	0.353	0.057	2.94
6	1.981	1.271	0.305	13:04:44	40.00	95.12	22.890	0.196	0.196	1.000	0.388	0.072	3.69
7	2.286	1.360	0.305	13:05:42	40.00	95.12	20.565	0.193	0.193	1.000	0.415	0.079	4.09
8	2.591	1.415	0.305	13:06:40	40.00	95.12	18.215	0.209	0.209	1.000	0.431	0.090	4.65
9	2.896	1.423	0.304	13:07:37	40.00	95.12	11.964	0.239	0.239	1.000	0.433	0.104	5.35
10	3.200	1.422	0.304	13:09:04	40.00	92.68	17.191	0.263	0.263	1.000	0.433	0.114	5.88
11	3.505	1.425	0.305	13:10:10	40.00	95.12	13.990	0.277	0.277	1.000	0.434	0.120	6.21
12	3.810	1.432	0.305	13:11:13	40.00	90.24	10.757	0.291	0.291	1.000	0.437	0.127	6.56
13	4.115	1.444	0.305	13:12:15	40.00	95.12	11.662	0.290	0.290	1.000	0.440	0.128	6.61
14	4.420	1.447	0.304	13:13:47	40.00	92.68	10.123	0.310	0.310	1.000	0.441	0.137	7.05
15	4.724	1.442	0.304	13:14:51	40.00	92.68	12.244	0.327	0.327	1.000	0.439	0.143	7.40
16	5.029	1.406	0.305	13:15:46	39.00	95.00	9.863	0.320	0.320	1.000	0.429	0.136	7.03
17	5.334	1.311	0.305	13:16:47	39.00	92.50	11.480	0.333	0.333	1.000	0.400	0.134	6.90
18	5.639	1.224	0.305	13:17:47	40.00	95.12	10.042	0.321	0.321	1.000	0.373	0.121	6.24
19	5.944	1.038	0.304	13:18:57	40.00	95.12	15.301	0.319	0.319	1.000	0.316	0.103	5.34
20	6.248	0.826	0.304	13:19:56	40.00	95.12	15.862	0.321	0.321	1.000	0.251	0.081	4.17
21	6.553	0.624	0.610	13:21:23	39.00	95.00	20.735	0.281	0.281	1.000	0.381	0.108	5.56
22	7.468	0.000	0.457	13:22:02	0.00	95.00	0.000	0.000	0.000	1.000	0.000	0.000	0.00

Diagnostic Files
 Raw File: sxs_000r.000 Config. File: sxs_000r.mmt

Field Notes:
 [01/07/2005 12:55 AM] Transect 0 aborted due to inability to program instrument.

Start *WinRiver II SxS* and open a measurement file.

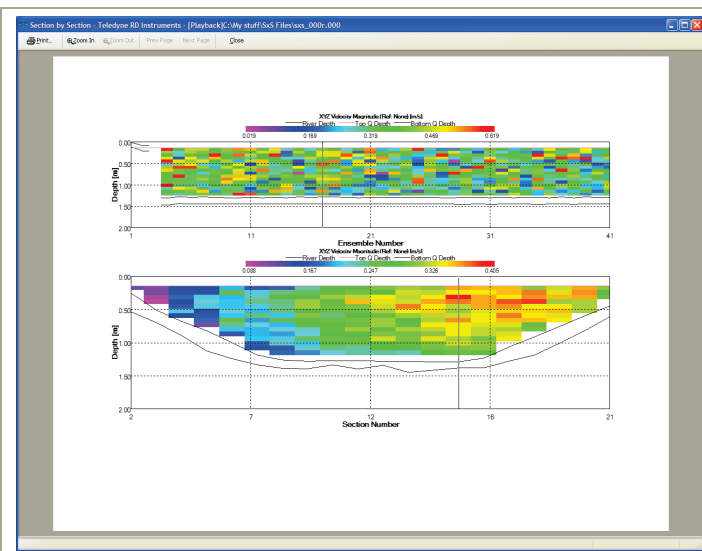
Playback / Reprocess the measurement (see [Reprocessing Data](#)).

Check/edit the Site Information by right-clicking on **Site Information** in the Section Project Control window and selecting **Site Wizard**.

On the **File** menu, click **Print Preview Q Measurement Summary**.

On the **Q Measurements** screen, click **Print** to print a copy of the discharge summary.

Print a Plot or Display



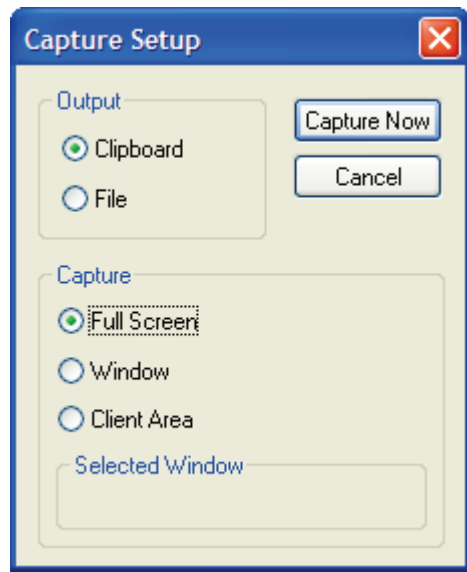
Click on the plot/display to be printed. The title bar will be highlighted.

On the **File** menu, click **Print Setup**. Select the desired setting and printer.

On the **File** menu click **Print Preview**. If the display is acceptable, click **Print**.

 Choose **Landscape** for contour plots.

Screen Captures



If you would like to save a plot/display for use in other documents, do the following.

Click on the plot/display to be saved. The title bar will be highlighted.

On the **Configure** menu, select **Capture Setup**. This will bring up the **Capture Setup** dialog box.

Select **Clipboard** or **File**.

Select **Full Screen**, **Window**, or **Client Area**. The **Client Area** will include all parts of the graph/plot except the title bars.

Click **Capture Now**. If you selected **File**, name the file and click **Save**.

Click **Cancel** to exit this dialog box.

Water Profiling Modes

This section explains all of the water-profiling modes available for the WorkHorse Rio Grande, Mariner ADCPs, and WorkHorse ADCPs with the high-resolution water profiling upgrade installed. For StreamPro systems, an additional Water Mode 13 is available. For each mode, we provide a general description, an explanation of the best place to use this mode, specifics about the mode, and any setup considerations. Use Table 1 as a guide for choosing the appropriate mode for your water flow conditions.

Typically if the flow is slow and the depth is shallow you would first try Mode 11. If the flow were too fast or turbulent for Mode 11 you would use Mode 12 if fitted. If Mode 11 or Mode 12 is not suitable then Mode 1 will work in all but the most extreme situations. Mode 5 and Mode 8 are still included for backward compatibility and for users who are familiar and satisfied with their performance.

Table 1: Recommended River Water Profiling Modes

	Mode 1	Mode 12	Mode 11	Mode 13	Mode 5	Mode 8
Typical application	Fast water of all depths. Rough and dynamic situations. Good in streams too fast or deep for modes 5, 8 & 11 or where Mode 12 has problems.	Fast water of all depths. Good in streams too fast or deep for modes 5, 8 & 11. Good for deep, slow water. See Note 1.	Slow, shallow streams with velocities < 1.0 m/sec (depth dependant) with low shear and/or turbulence.	Slow, shallow streams where the Maximum Stream Depth is ≤ 1.0 meters and the Maximum Stream Velocity is < 0.25 m/s	Slow, shallow streams with velocities < 0.5 m/sec with low shear and/or turbulence.	Shallow streams with velocities < 1 m/sec and with moderate shear (rough bed) and/or turbulence.
Minimum recommended cell size (meters)	0.50* 0.25	0.25* 0.10	0.10 0.05	StreamPro ADCPs Only See Low Noise Mode/Water Mode 13	0.10 0.10	0.10 0.10
Recommended Cell Size (meters)	0.50 0.25	0.25 0.10	0.25 0.05		0.25 0.10	0.25 0.10
Single ping standard deviation (cm/s) (using rec. cell size)	13.62 13.64	6.24 6.95	0.74 1.34		0.33 0.44	3.34 5.15
First range cell (meters)	0.97 0.51	0.73 0.26	0.49 0.09		0.49 0.14	0.49 0.14
Minimum profiling range (meters) Bottom Mode 5	1.7 1.0	1.7 1.0	1.6 0.9		1.6 0.9	0.9 0.6
Minimum profiling range (meters) Bottom Mode 7	NA 0.7	NA 0.5	NA 0.3		NA 0.7	NA 0.3
Maximum profiling range (meters)	73.1 19.55	68.29 15.82	<8.0 <4.0		<8.0 <4.0	<8.0 <4.0
Maximum relative velocity (m/s)	10 m/sec	10 m/sec	1 m/sec (Depth Dependent)		0.5 m/sec	1 m/sec

* 600 kHz values are in **bold** font, and 1200 kHz values are in regular font.

Specifications are for 25 cm blank (600 kHz), 5cm Blank (1200 kHz), 10° C temperature, and 0.0 salinity.

Note 1. Mode 12 table assumes 20 sub-pings (WO20,4).



NOTE. The ranges in Table 1 are measured from the transducer face. Add the transducer depth to determine the actual minimum and maximum profiling depths.



NOTE. Maximum range depends on water temperature and depth cell size. Use *PlanADCP* to compute the maximum range for a particular ADCP set-up and water temperature. The standard deviation of modes 5, 8 and 11 varies with water speed, boat speed, bed-form roughness, channel depth, and turbulence.

Commonly Used ADCP Commands

The commands shown in Table 2 (these are only a small percentage of those available) can all be found with detailed descriptions in the WorkHorse Commands and Output Data Format guide, however below are some of the more commonly used commands.

Table 2: Commonly Used ADCP Commands

Command	Description
BX	Max depth in decimeters (meters x 10) that the ADCP will look for the bottom - set this greater than your deepest depth.
WS	Depth cell size (cm)
WN	Number of depth cells (WS x WN = BX x 10)
WF	Blanking distance (cm)
WV	Ambiguity velocity (160 to 200 is usually good) (approximately water velocity in cm/sec)
WM	Water mode 1, 5, 8, 11 or 12 (1200 Rio Grande / Zed Head)
WP	Water pings
BP	Bottom pings
CL	Sleep Between Pings (CL0 = Do Not Sleep, CL1 = Sleep Between Pings)

Commonly Used BBTalk Commands

Once *BBTalk* is running and connected to the ADCP, commands can be entered to interrogate the unit. The commands shown in Table 3 (these are only a small percentage of those available) can all be found with detailed descriptions in the WorkHorse Commands and Output Data Format guide.

Table 3: Commonly Used BBTalk Commands

Command	Description
CR1	Parameters set to FACTORY Settings
CK	Parameters saved as USER defaults
PS0	Basic Instrument Information including Serial Number, Frequency, Firmware, and Board Serial Numbers
PT200	Built in Self Test
CB	Change baud rate
PC1	Beam Continuity Test - Rub beams when asked to do so. ADCP must be in air to run this test.
PC2	Position, direction, temperature etc - Press any key to quit
PA	Pre-deployment Tests
?	List possible inputs. If added at the end of a command (i.e. CB?), it will list the command setting.
CS	Start Pinging. Once ADCP is pinging, send a Break to stop.
= = =	Send Break (Radio Modem)
End	Send Break (Direct Cable)

WinRiver II SxS Shortcut Keys

Key	Description
F1	Help
F4	Start/Stop Pinging
F5	Start/Stop Measurement
F7	Properties
F8	Toggle Bank
F9	Toggle Ensemble Header Tabular view
F11	Toggle Detailed Discharge/Composite Tabular view
F12	Toggle Discharge Summary Tabular view
Ctrl-A	Set as Active Configuration (when Measurement Control window selected)
Ctrl-B	Reference - Bottom Track
Ctrl-C	Copy
Ctrl-D	Toggle Acquire Control window
Ctrl-E	Close Measurement File
Ctrl-F	Create Measurement from Data Files
Ctrl-F2	Apply to All Active Configurations
Ctrl-F3	Toggle Measurement Management window (when selected)
Ctrl-F5	Reprocess Measurement
Ctrl-F7	Properties (when window or file selected)
Ctrl-F8	Data Selection (when Profile or Contour plot selected)
Ctrl-U	Duplicate Configuration node (when Measurement Control window selected)
Ctrl-T	Delete Configuration node (when Measurement Control window selected)
Ctrl-T	Export to CVS (When Discharge Summary selected)
Ctrl-J	Rename Configuration node (when Measurement Control window selected)
Ctrl-R	Reset Configuration node Properties (when Measurement Control window selected)
Ctrl-I	Site Wizard
Ctrl-K	Add Note (when Measurement Control window selected)
Ctrl-L	Lock/Unlock Measurement File
Ctrl-M	View Command Log
Ctrl-N	Reference - None
Ctrl-O	Open measurement file
Ctrl-P	Print
Ctrl-S	Save Measurement File
Ctrl-W	New Measurement
Shift-F4	Set ADCP Clock
Shift-F8	Execute RGTest
Shift-F9	Execute Pressure Sensor Test
Shift-F10	Execute Compass Calibration
Minus	Previous Ensemble (Playback)
Space	Next Ensemble (Playback)
Ctrl-Minus	Previous Section (Playback)
Ctrl-Space	Next Section (Playback)
Home	First Ensemble (Playback)
End	Last Ensemble (Playback)

Where to Find More Information

Congratulations! You have completed the Quick Start Guide. For more detailed information, see the following sections in the *WinRiver II Section-By-Section* User's Guide.

Using the Measurement Wizard – This section covers how to create a configuration node using the **Measurement Wizard** and a detailed explanation of the configuration node settings.

Acquiring Discharge Data – This section is a more detailed explanation of the steps covered in the Quick Start Guide.

Post-Processing of Discharge Data – This section covers data playback, processing, and screen captures.

NOTES